

GENERAL SCIENCE GRADE 10: THE PERIODIC TABLE

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.1 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	General Science
Strand	Strand 2.0: Matter and Chemical Reactions
Sub-Strand	Sub-Strand 2.1: The Periodic Table
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Classification of the first twenty elements of the periodic table into groups and periods – Electron configuration of atoms of elements (atomic number, mass number, electron arrangement) – Stability and electron affinity of atoms — how elements attain stable configurations (cations and anions) – The periodic law and the order adopted in developing the periodic table – Unreactive nature of noble gases – Dot (•) and cross (×) diagrams illustrating ion formation for the first 20 elements – Valency and oxidation number of elements and radicals (ammonium ion, sulphate, nitrate, hydroxide, carbonate, phosphate, hydrogen carbonate) – Chemical formulae of common compounds using valencies of elements and radicals – Writing and balancing chemical equations of chemical reactions – Project: Modelling electron configuration of any five elements (1–20) using locally available materials – Project: Chart of common substances made from elements and radicals (names and formulae)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - examine electron arrangement of an atom of an element, - classify elements into groups and periods of the periodic table, - explain the stability and electron affinity of atoms of elements, - describe ion formation for the first 20 elements of the periodic table, - formulate chemical formulae of common compounds, - write balanced equations of chemical reactions, - appreciate the importance of the structure of the atom in the development of the periodic table.
Core Competencies	– Critical Thinking and Problem Solving: as the learner sorts and arranges the first twenty elements of the periodic table according to their similarities in electron arrangement and deduces patterns of groups and periods. – Creativity and Imagination: as the learner models the electron configuration of any five elements of the periodic table (1–20) using locally available materials such as maize cobs, bottle tops, or wire. – Digital Literacy: as the learner watches animations and simulations on ion formation and atomicity of gases using digital devices, and researches the periodic law using online or print resources.

Core Values	– Respect: as the learner recognises and respects the input of every group member when discussing how to determine the valency and oxidation numbers of elements and radicals. – Unity: as the learner takes turns playing roles when drawing dot (•) and cross (×) diagrams to illustrate ion formation for atoms of elements of the periodic table. – Responsibility: as the learner uses digital devices appropriately to watch simulations and animations on ion formation and handles laboratory or modelling materials with care.
Science & Engineering Practices	– Developing and Using Models: as the learner draws dot-and-cross diagrams and constructs physical models of electron configurations of elements using locally available materials to represent atomic structure. – Analysing and Interpreting Data: as the learner examines electron arrangements of the first 20 elements and identifies patterns that justify their classification into groups and periods. – Constructing Explanations: as the learner explains why atoms of elements gain or lose electrons to achieve stability, linking electron affinity to ion formation and the formation of cations and anions. – Using Mathematics and Computational Thinking: as the learner applies valency rules and oxidation numbers to formulate chemical formulae and balance chemical equations systematically.
Pertinent & Contemporary Issues (PCIs)	– Life Skills (Problem Solving Skills): as the learner arranges the first twenty elements of the periodic table into groups and periods, applying logical reasoning to identify patterns. – Financial Literacy: as the learner uses locally available materials (bottle tops, maize seeds, wire, clay) as a way of minimising expenses when modelling the electron configuration of any five elements of the periodic table.
Career Connections	– Pharmacy and Medicine: understanding ion formation and valency is foundational for drug formulation, electrolyte balance in patient care, and reading chemical formulae on medicine labels — relevant in Kenya's growing healthcare sector. – Chemical Engineering and Manufacturing: classifying elements and balancing equations underpins industrial chemical production at facilities such as Athi River Mining (ARM Cement) and East African Portland Cement in Kenya. – Agriculture and Agro-chemistry: knowledge of radicals such as nitrate, phosphate and ammonium ion directly connects to understanding fertilisers (CAN, DAP, Urea) used by Kenyan farmers in the Rift Valley and Central Kenya. – Mining and Geology: the periodic table is the foundation for understanding mineral composition and ore processing in Kenya's mining sector (e.g., soda ash at Magadi, titanium in Kwale County).
Focus for Lessons	This unit develops learners' understanding of how the electron arrangement of the first twenty elements provides the organising principle behind the periodic table — connecting atomic structure to observable chemical behaviour. Using a phenomenon-driven, model-building approach rooted in the Kenyan context, learners move from reviewing atomic structure to classifying elements, explaining ion formation, and ultimately writing and balancing chemical equations. The unit places the abstract architecture of the atom in the service of explaining everyday Kenyan experiences, from why salt (NaCl) dissolves in water for ugali preparation to why fertiliser bags carry specific chemical formulae.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why do the fertiliser bags on Kenyan farms carry such specific and different chemical formulae — and what does the inner structure of an atom have to do with it? KICD KEY INQUIRY QUESTIONS: 1. Why is electron arrangement significant? 2. How are elements arranged in the periodic table?
Anchoring Phenomenon	A Grade 10 learner in Kericho notices that the fertiliser bag used on her family's tea farm is labelled "CAN — Calcium Ammonium Nitrate, $\text{Ca}(\text{NO}_3)_2 \cdot \text{NH}_4\text{NO}_3$." She also notices that a different bag labelled "DAP — Di-ammonium Phosphate, $(\text{NH}_4)_2\text{HPO}_4$" looks almost identical but is used at a different stage of growth. Both bags contain white crystalline solids that dissolve easily in water,

	yet the plants respond differently to each. She wonders: why do these two compounds have such different formulas and different effects, even though both contain nitrogen? She knows from Junior School that matter is made of atoms — but how does the inner structure of an atom determine what formula a compound gets, and why do certain elements always combine in fixed ratios? This is puzzling because from the outside, both compounds look similar, yet their chemical identities — encoded in their formulae — are completely different, and those formulae seem to follow a hidden pattern she cannot yet explain.
Storyline Thread	Lesson 1 — Anchoring the Phenomenon & Reviewing Atomic Structure: Students examine real fertiliser bags (CAN and DAP) and the driving question is posed. Learners review atomic number, mass number, and electron arrangement of selected elements (H, C, N, O, Na, Cl, Ca) through guided activities, connecting prior Junior School knowledge to the new context. Students make initial predictions on the DQB about why the formulae differ. Lesson 2 — Electron Configuration and the Periodic Law: Learners carry out a card-sorting activity arranging the first 20 elements by their electron arrangements. They discover the repeating pattern of outer-shell electrons and discuss the periodic law with peers. The connection between electron shells and the rows (periods) and columns (groups) of the periodic table is established. Learners annotate a blank periodic table grid. Lesson 3 — Stability, Electron Affinity, and the Noble Gas Rule: Students investigate why noble gases are unreactive by examining their full outer shells. Learners discuss how other atoms lose or gain electrons to attain stability, introducing the concepts of electron affinity, cations, and anions. The anchoring phenomenon is revisited: why does calcium (Group 2) form Ca^{2+} and nitrogen (Group 5) form N^{3-}? Lesson 4 — Ion Formation: Dot-and-Cross Diagrams: Learners draw dot (•) and cross (×) diagrams to illustrate ion formation for selected elements from the first 20 (e.g., Na^+, Mg^{2+}, Al^{3+}, Cl^-, O^{2-}). They watch animations/simulations on ion formation using available digital devices. Groups present their diagrams to the class, tying ion charges back to the periodic table group number. Lesson 5 — Valency, Oxidation Numbers, and Radicals: Learners derive valency from group position and ion charge. The teacher introduces oxidation numbers and key radicals verbatim from KICD: ammonium ion (NH_4^+), sulphate (SO_4^{2-}), nitrate (NO_3^-), hydroxide (OH^-), carbonate (CO_3^{2-}), phosphate (PO_4^{3-}), hydrogen carbonate (HCO_3^-). Students practise determining valency and oxidation numbers through worked examples rooted in Kenyan compounds (e.g., urea, calcium nitrate). Lesson 6 — Writing Chemical Formulae of Common Compounds: Using valencies of elements and radicals established in Lesson 5, learners systematically write chemical formulae for common compounds (NaCl, MgO, CaCO_3, Al_2O_3, NH_4NO_3, $(\text{NH}_4)_2\text{HPO}_4$). They decode the fertiliser bag formulae from the anchoring phenomenon, resolving part of the driving question. A class chart of common substances with names and formulae is begun (Project 2). Lesson 7 — Balancing Chemical Equations: Learners apply valency knowledge to write and balance chemical equations for reactions involving the first 20 elements (e.g., formation of sodium chloride, magnesium oxide, calcium nitrate from constituent elements/compounds). Step-by-step balancing strategies are practised in pairs, with peer checking. Students connect balanced equations back to the conserved ratios seen in fertiliser formulae. Lesson 8 — Model Building, Project Showcase & DQB Resolution: Learners present their electron-configuration models (Project 1, using locally available materials — bottle tops, wire, clay) and their completed compound charts (Project 2). The class revisits the DQB, resolving the driving question by explaining in full how electron arrangement → valency → chemical formula → the labels on Kenyan fertiliser bags. A summative reflection links the unit to careers in agriculture, pharmacy, and chemical manufacturing in

	Kenya.
Supporting Phenomena	– When a Nairobi hardware vendor cuts sodium metal (stored under oil) and drops a tiny piece into water, it fizzes violently, skids across the surface, and may catch fire — yet the sodium atom has only one more electron than the noble gas neon. Why does that single extra electron make sodium so reactive while neon is completely inert? – A student at a school in Kisumu observes that rust (iron oxide, Fe_2O_3) forms on an iron gate near Lake Victoria far faster than on an aluminium window frame next to it, even though both metals are exposed to the same humid air. Why do different metals react differently with oxygen, and can the periodic table predict this? – A cook in Mombasa dissolves common table salt (NaCl) — made from the metal sodium and the poisonous gas chlorine — into water to make a perfectly safe brine for pickling fish. How can two dangerous elements combine to form something harmless, and what happens at the atomic level during that combination?

LESSON 1 (40 minutes): Anchoring the Phenomenon and Reviewing Atomic Structure

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners will anchor the sub-strand phenomenon by examining real Kenyan fertiliser bags and will connect prior Junior School knowledge of atomic structure to the new driving question about why chemical formulae differ.
Knowledge	Learners can state the atomic number and mass number of selected elements (H, C, N, O, Na, Cl, Ca); learners can write the electron arrangement of these elements using the 2-8-8 shell model; learners can identify protons, neutrons, and electrons from atomic data.
Skills	Observing and recording information from real fertiliser bag labels; retrieving and applying prior knowledge of atomic structure; making and articulating initial predictions; constructing a simple electron-arrangement notation.
Attitudes	Curiosity about why matter behaves differently despite looking similar; appreciation of how chemistry connects to everyday Kenyan farming practice; willingness to share initial ideas even when uncertain.
Key Inquiry Question	Why is electron arrangement significant? (Key Inquiry Question 1 — introduced at the opening level; full resolution is built across the sub-strand.)
Purpose in Storyline	This is the opening lesson of the sub-strand. It launches the anchoring phenomenon, activates prior knowledge of atomic structure as the foundation for all subsequent learning, and creates the Driving Question Board that will be revisited and resolved in Lesson 8.
Safety Notes	No hazardous chemicals are handled. Fertiliser bags are empty and clean. Remind learners to wash hands after handling any packaging material. Ensure the classroom is well ventilated.

B. LESSON OVERVIEW

The teacher opens by displaying two empty, clean fertiliser bags — CAN (Calcium Ammonium Nitrate, $\text{Ca}(\text{NO}_3)_2 \text{NH}_4\text{NO}_3$) and DAP (Di-ammonium Phosphate, $(\text{NH}_4)_2\text{HPO}_4$) — sourced from a local agro-dealer in the Kericho tea-farming region. Learners observe the labels, note the formulae, and articulate what puzzles them. The driving question is posed and the Driving Question Board is created. Learners then work in pairs on a guided card-matching activity to review atomic number, mass number, and electron arrangement for seven key elements (H, C, N, O, Na, Cl, Ca), connecting Junior School knowledge to the phenomenon. The lesson closes with learners recording initial predictions on sticky notes placed on the DQB.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Video: Isotopes and Atomic Mass Source: PhET Interactive Simulations	Before seeing the fertiliser bags, learners respond individually in their exercise books to a brief prompt written on the board: 'Think about the food crops or tea on a farm near you. What do you think farmers add to the soil to help	Teacher writes the prompt on the board before learners enter. As learners think silently, teacher circulates and reads responses without commenting. After 2 minutes teacher says: 'Share your thinking with the person next to	Think-Pair-Share activates prior informal knowledge about farming and chemistry, making learners aware of what they already know and what they are uncertain about before the phenomenon is	Teacher listens to pair conversations and reads written sentences to gauge the range of prior knowledge about compounds and farming — no grade is assigned; information is used to calibrate pacing in the Observe

(English) Search ARES for similar videos HTML: Dalton's Atomic Theory (Flexbook) Source: CK-12 Search ARES for similar readings	plants grow, and why might different crops or growth stages need different substances?' After 2 minutes of silent thinking, learners share ideas with a neighbour using a Think-Pair-Share structure. Selected pairs share with the class. Learners then write one sentence predicting: 'I think the formula of a fertiliser compound depends on _____ because _____.'	you — you have 90 seconds.' [WAIT TIME: 90 seconds of pair talk — teacher listens but does not intervene.] Teacher then says: 'I heard some interesting ideas. Who would like to share — and it is fine to say you are not sure yet?' Teacher takes three to four responses, writes key words (e.g., nutrients, minerals, chemicals, growth) on the board without evaluating. Teacher closes with: 'You have raised excellent starting ideas. Now let us look at something from a real Kenyan farm that will sharpen our thinking.' [WAIT TIME after this statement: 10 seconds of silence to build anticipation before revealing the bags.]	introduced.	Phase.
Observe Phase VIDEO: Video: Isotopes and Atomic Mass Source: PhET Interactive Simulations (English) Search ARES for similar videos HTML: Dalton's Atomic Theory (Flexbook) Source: CK-12 Search ARES for similar readings	The teacher places one CAN bag and one DAP bag on the front demonstration table where all learners can see them. Learners move into groups of four. Each group receives a printed A4 observation sheet containing: (a) a photograph of each bag label with the formula clearly visible, (b) three observation prompts: 'What do you see that is the SAME about the two bags?', 'What do you see that is DIFFERENT?', 'What do you find PUZZLING?' Groups spend 5 minutes writing observations. The teacher then reads aloud the anchoring scenario: 'A Grade 10 learner in Kericho notices that the fertiliser bag on her family tea farm is labelled CAN — $\text{Ca}(\text{NO}_3)_2$ NH_4NO_3 — and another bag labelled DAP — $(\text{NH}_4)_2\text{HPO}_4$ — looks almost identical but is used at a different stage of growth. Both are white crystals that dissolve easily in water, yet the plants	Teacher holds up each bag and says: 'Take 30 seconds to just look — do not write yet. Use your eyes.' [WAIT TIME: 30 seconds of silent looking.] Teacher then says: 'Now fill in your observation sheet with your group. I will give you 5 minutes.' [WAIT TIME: 5 minutes of group work — teacher circulates, crouches to group level, reads sheets, and asks probing questions such as: 'What exactly on the label caught your attention?' and 'You wrote they look similar — what specifically is similar?' without giving answers.] After 5 minutes teacher reads the scenario in a calm, storytelling voice. Teacher then says: 'Now add anything new you notice or wonder to your sheet. You have 2 more minutes.' [WAIT TIME: 2 minutes.] Teacher selects two groups to share one observation each and one puzzling question each. Teacher records the puzzling questions on the board	Structured observation sheet with same-different-puzzling prompts focuses learner attention on the phenomenon before explanation is offered, building the need-to-know that drives genuine inquiry.	Teacher reads group observation sheets during circulation to identify: whether learners notice that both formulae contain N (nitrogen) but differ in other elements and ratios; whether learners connect the formulae to atomic symbols from Junior School. Groups that do not notice the nitrogen connection are prompted with: 'Look carefully at each formula — do you see any symbols appearing in both?'

	respond differently. Both contain nitrogen. She wonders: why do these two compounds have such different formulae and different effects?' Learners add new observations and questions to their sheets after hearing the scenario.	under the heading 'We are puzzled by...'		
Explain Phase  VIDEO: Video: Isotopes and Atomic Mass Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: Dalton's Atomic Theory (Flexbook) Source: CK-12  Search ARES for similar readings	The teacher transitions to a guided review of atomic structure, framing it explicitly as the tool needed to begin answering the driving question. Each pair receives a set of seven element cards (pre-printed, one per element: H, C, N, O, Na, Cl, Ca). Each card shows: element name, symbol, atomic number, and mass number. Pairs use the information on the cards to: (1) calculate the number of protons, neutrons, and electrons for each element; (2) write the electron arrangement using the notation shells e.g. 2.8.8; (3) draw a simple Bohr-model diagram for N and Ca. The teacher models the process live on the board using oxygen (O, atomic number 8, mass number 16) before pairs begin: 'Protons equal atomic number, so oxygen has 8 protons. In a neutral atom, electrons equal protons, so 8 electrons. Neutrons equal mass number minus atomic number, so 16 minus 8 equals 8 neutrons. Electron arrangement: first shell holds up to 2, so 2 electrons there; second shell holds up to 8, and we have 6 remaining, so arrangement is 2.6.' Pairs work for 8 minutes. Teacher facilitates a whole-class check by asking individuals to call out their answers while the teacher writes on the board, pausing for peer correction.	Before distributing cards, teacher says: 'You learned about atomic structure in Junior School. We are going to use that knowledge now because it is the key to unlocking why these fertiliser formulae look the way they do.' Teacher models oxygen clearly and slowly, narrating each step. Teacher then says: 'You have 8 minutes to complete the table for all seven elements. If you are unsure, look at the model on the board.' [WAIT TIME: 8 minutes of pair work — teacher circulates, noting common errors such as confusing mass number with neutron count.] During class check teacher asks: 'What is the electron arrangement of nitrogen?' [WAIT TIME: 5 seconds.] If no response: 'Nitrogen has atomic number 7 — how many electrons is that, and how do we fill the shells?' Teacher ensures electron arrangement of Ca (2.8.8.2) is explicitly recorded and says: 'Notice that calcium has 2 electrons in its outermost shell. Keep that number in mind — it will matter a great deal in the lessons ahead.' Teacher connects back to phenomenon: 'Calcium is in CAN fertiliser — $\text{Ca}(\text{NO}_3)_2$. Why might calcium always appear with that 2 in the formula? That is part of what we will explain across this sub-strand.'	Worked example followed by structured pair practice (I do — We do — You do) ensures all learners retrieve and correctly apply the electron arrangement model before it is extended in later lessons. Explicit connection of Ca electron arrangement to the fertiliser formula begins to thread the phenomenon through the content.	Teacher scans completed tables during circulation. Key checkpoint: can the learner correctly derive electron arrangement from atomic number alone? Learners who confuse atomic number and mass number are redirected with: 'Which number tells us how many protons — and therefore how many electrons in a neutral atom?' Exit data from this phase informs the depth of review needed at the start of Lesson 2.
Driving Question	The teacher formally poses the Driving Question by reading it	Teacher says: 'This question will guide everything we do in this sub-	The DQB externalises learner thinking, makes questions public	Teacher reads all sticky notes before Lesson 2. Questions are

Board (DQB) Creation  VIDEO: DNA as the "transforming principle" Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Dalton's Atomic Theory (Flexbook) Source: CK-12  Search ARES for similar readings	aloud and writing it on a large dedicated board or wall space that will remain visible for all eight lessons: 'Why do the fertiliser bags on Kenyan farms carry such specific and different chemical formulae — and what does the inner structure of an atom have to do with it?' Each learner receives two sticky notes. On the first (yellow) sticky note, learners write one question they have after today's lesson — something they genuinely do not yet understand. On the second (green) sticky note, learners write their current best prediction in one sentence answering the driving question. Learners come forward and place both sticky notes on the DQB. The teacher reads a selection of questions aloud and groups similar ones together on the board with a marker, creating clusters labelled: 'About atoms', 'About formulae', 'About farming'. Learners copy the driving question into their exercise books and underline it.	strand. You will not be able to fully answer it today — and that is completely expected and fine. Your job right now is to be honest about what you are wondering.' Teacher gives learners 3 minutes to write both sticky notes in silence. [WAIT TIME: 3 minutes of silent individual writing.] As learners place notes, teacher reads two or three aloud: 'Someone has asked: why does calcium have a 2 after it in the formula? That is a brilliant question and we will answer it. Someone else asks: what does NH₄ mean? Another excellent question.' Teacher then says: 'Notice that your questions are not random — they cluster around atoms, around formulae, and around what happens on real farms. That is exactly the structure of this sub-strand.' Teacher points to the DQB and says: 'This board belongs to you. We will return to it in every lesson and in Lesson 8 you will answer it completely.'	and legitimate, creates shared ownership of the inquiry, and provides the teacher with formative data on conceptual gaps. Clustering questions helps learners see the coherence of the sub-strand before content is taught.	sorted into three categories: (1) questions answerable by end of Lesson 2-3 (electron arrangement, periodicity); (2) questions answerable by Lesson 5-6 (valency, formulae); (3) questions requiring full sub-strand resolution. This informs differentiation planning.
Model Building Phase  VIDEO: Video: Isotopes and Atomic Mass Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: Dalton's Atomic Theory (Flexbook) Source: CK-12  Search ARES for similar readings	In the final segment, each learner individually constructs a simple electron-arrangement notation model for the seven elements studied today. Using their exercise books, learners draw concentric circle (Bohr) diagrams for at least three elements of their choice from the seven, labelling: nucleus (protons and neutrons), electron shells, and number of outer-shell electrons. Learners then write a brief reflection (two to three sentences) answering: 'What did I know before today? What am I now noticing or thinking differently? What am I still wondering?' The teacher collects or photographs these reflections	Teacher says: 'You have 4 minutes to draw your three Bohr diagrams. Choose elements that feel important to you — maybe elements you noticed in the fertiliser formulae.' [WAIT TIME: 4 minutes of individual silent work.] Teacher circulates and stamps or ticks completed diagrams to acknowledge effort. Teacher then says: 'Now write your reflection — two or three sentences, honest thinking, no wrong answers.' [WAIT TIME: 2 minutes.] Teacher invites one volunteer to read their reflection aloud and responds: 'Thank you — you have captured exactly the kind of thinking a scientist holds: knowing some	Individual model-drawing consolidates the reviewed content in a concrete visual form that learners will physically revisit and extend in later lessons (Project 1 begins formally in Lesson 2 but is seeded here). The written reflection activates metacognition and gives the teacher a window into each learners conceptual state.	Collected exercise-book reflections reveal which learners have a solid atomic-structure foundation (ready to engage deeply with periodicity in Lesson 2) and which learners have significant gaps (require brief targeted review at the start of Lesson 2 or a differentiated support card). Bohr diagrams are checked for correct shell filling.

	before the next lesson. A brief whole-class share of one or two reflections closes the lesson. The teacher previews Lesson 2 by saying: 'In our next lesson you will sort all first 20 elements by their electron arrangements and discover a hidden pattern that explains why the periodic table looks the way it does — and that pattern is the key to the fertiliser formulae.'	things, noticing new things, and staying curious about the rest.' Teacher closes with the Lesson 2 preview, pointing to the DQB: 'Your questions up there will start getting answers next lesson.'		
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D. TEACHER REFLECTION

After the lesson consider: Did all learners engage with the phenomenon as genuinely puzzling, or did some treat it as a routine content introduction? Were the sticky-note questions substantive and science-focused, or superficial? Did the electron-arrangement review reveal gaps that will slow Lesson 2? Were there learners who could not recall Junior School content and need a differentiated support card for Lesson 2? How well did the Kericho-based farming context resonate — did rural and urban learners connect equally? Note which DQB question clusters are largest, as these indicate where the most instructional time may be needed across the sub-strand.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Two fertiliser bags used on Kenyan farms — CAN ($\text{Ca}(\text{NO}_3)_2 \text{NH}_4\text{NO}_3$) and DAP ($(\text{NH}_4)_2\text{HPO}_4$) — look similar on the outside as white crystalline solids that dissolve in water, yet they carry completely different chemical formulae and affect plant growth differently. Both contain nitrogen, but their formulae differ in the elements present and the ratios in which those elements appear.
What did I learn?	Atoms of each element have a fixed number of protons (atomic number) and a characteristic electron arrangement described using the 2-8-8 shell model. The elements in the fertiliser formulae — H, N, O, Ca, P — each have a specific atomic number and a specific pattern of electrons distributed across their shells. The number of electrons in the outermost shell appears to be particularly important.
How does this explain the phenomenon?	We cannot yet fully explain why the formulae are different, but we now know that the answer is connected to the electron arrangements of the atoms involved. The number of outer-shell electrons of calcium (2.8.8.2 — two outer electrons) somehow relates to the 2 appearing in $\text{Ca}(\text{NO}_3)_2$. Understanding this connection between electron arrangement and formula will be built across the sub-strand.

LESSON 2 (80 minutes (2 × 40-minute lessons)): Electron Configuration and the Periodic Law: Finding the Hidden Pattern

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners will discover that the repeating pattern of outer-shell electrons is the hidden rule that determines how elements are arranged in the periodic table — and that this same rule underlies why CAN and DAP have different chemical formulae and different effects on tea plants.
Knowledge	Learners examine the electron arrangement of the first 20 elements, classify elements into periods (rows) and groups (columns) of the periodic table, and explain the periodic law as the repeating pattern of outer-shell (valence) electrons.
Skills	Learners carry out a card-sorting investigation (SEP: Analysing and Interpreting Data), construct arguments from evidence about the repeating pattern (SEP: Engaging in Argument from Evidence), and annotate a blank periodic table grid using data from their card sort.
Attitudes	Learners appreciate that the periodic table is not an arbitrary list but a deeply ordered structure whose logic connects directly to the compounds that Kenyan farmers use every season, fostering respect for the contributions of scientists who discovered this order.
Key Inquiry Question	Why is electron arrangement significant? How are elements arranged in the periodic table?
Purpose in Storyline	In Lesson 1 learners reviewed atomic structure. Now in Lesson 2 they gather the first major piece of evidence: the repeating pattern of valence electrons across the first 20 elements. This pattern — the periodic law — is the hidden logic behind why elements bond in fixed ratios, which will ultimately explain why CAN is $\text{Ca}(\text{NO}_3)_2 \cdot \text{NH}_4\text{NO}_3$ and DAP is $(\text{NH}_4)_2\text{HPO}_4$. Without understanding the repeating valence-electron pattern, learners cannot yet explain why calcium always forms Ca^{2+} or why nitrogen forms NO_3^- .
Safety Notes	This lesson is card-based and paper-based — no chemicals or apparatus with hazard potential are used. Ensure learners handle printed cards carefully and that the classroom floor is clear of tripping hazards during the gallery-walk phase.

B. LESSON OVERVIEW

Learners begin by revisiting the CAN and DAP fertiliser bags from the Kericho phenomenon and making a focused prediction: if you listed the first 20 elements in order of atomic number and wrote down how many electrons are in their outermost shell, what pattern — if any — would you see? They then carry out a structured card-sorting activity (the core evidence-gathering experience), arranging 20 element cards by electron configuration data and hunting for repeating patterns. A peer-discussion protocol surfaces the periodic law. Learners annotate a blank periodic table grid, placing elements into correct periods and groups, and record updated thinking on the Driving Question Board. The lesson closes with a model-revision task in which learners add a 'valence electron layer' to the atomic models they began in Lesson 1.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Quantum numbers	Learners individually read two data strips placed on their desks. Strip A shows the full electron arrangement of hydrogen (2,1	Distribute Strip A and Strip B (pre-printed on the ARES platform or photocopied). Say: 'Before we sort any cards, I want your honest	Prediction Journaling (written first, spoken second). By committing a prediction to paper before the card sort, learners activate prior	Circulate during the 4-minute writing period. Look for: (1) learners who correctly identify a repeating pattern in Strip A (e.g.,

Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Electron Cloud Atomic Model (Flexbook) Source: CK-12 Search ARES for similar readings	electrons — note: H is 1, Li is 2,1) through the first 10 elements. Strip B shows only the atomic number and name of elements 11–20, with the electron-arrangement column left blank. Each learner silently writes a 3-sentence prediction in their exercise book answering: 'If I continued filling in the outer-shell electron numbers for elements 11–20, what pattern do you think I would find — and why might that pattern matter for how a fertiliser like CAN gets its formula?' Learners are NOT shown a periodic table at this stage.	prediction — written down, in your own words. There is no wrong answer here; we are scientists making a claim before we collect evidence. You have 4 minutes of silent writing. Go.' Set a visible timer. After 4 minutes, cold-call 3 learners (aim for gender balance — e.g., one girl from row 1, one boy from row 3, one learner of your choice). For each response, ask: 'What evidence from Strip A led you to that prediction?' Allow 10 seconds of WAIT TIME after each question before accepting answers. Record the 3 predictions verbatim on the board under the heading 'Our Predictions — We Will Test These.'	knowledge from Lesson 1 (atomic structure) and create a personal cognitive stake in the upcoming evidence. Comparing written predictions to card-sort results later produces productive cognitive conflict — a powerful driver of conceptual change.	'the outer shell goes 1, 2, 3, 4 ... then seems to reset') — these learners are ready to lead the card sort; (2) learners who write only element names with no reference to electron numbers — flag these learners for targeted questioning during Phase 2. Do not correct misconceptions at this stage; note them on a clipboard for follow-up.
Observe Phase VIDEO: Quantum numbers Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Electron Cloud Atomic Model (Flexbook) Source: CK-12 Search ARES for similar readings	In groups of 4, learners receive a set of 20 pre-cut Element Cards (one per element, H to Ca). Each card shows: element name, symbol, atomic number, full electron arrangement (e.g., Na: 2,8,1), and a small blank box labelled 'outer-shell electrons.' Step 1 (5 min): Each group writes the number of outer-shell electrons in the blank box on every card. Step 2 (10 min): Groups arrange all 20 cards on their desk in a way that best reveals a pattern — they may arrange by atomic number, by outer-shell count, by total shells, or any combination they choose. They must be able to explain their arrangement. Step 3 (5 min): Each group records in their books: 'We arranged the cards by _____. The pattern we found is _____. This surprised / did not surprise us because _____.' Step 4 (5 min): Groups do a silent gallery walk — they move to the next group's desk, observe that group's arrangement, and write one	Before distributing cards, say: 'Your job right now is not to memorise — your job is to find the hidden pattern that Dmitri Mendeleev found in 1869 and that a Kenyan chemist today uses every time she reads a fertiliser formula. Look for something that repeats.' Circulate continuously. At the 8-minute mark, if any group has not yet discovered the repeating outer-shell pattern, crouch to desk level and ask: 'How many outer-shell electrons does lithium have? How many does sodium have? How many does potassium have? What do you notice?' Allow 10 seconds of WAIT TIME. Do NOT give the answer — respond to any answer with 'Interesting — what does that tell you about their column position?' During the gallery walk, position yourself at the desk of the group that produced the most unconventional arrangement and ask the visiting group: 'What is one thing this arrangement shows that	Card Sorting as Data Analysis (NGSS SEP 4: Analysing and Interpreting Data). Physical manipulation of cards externalises the reasoning process — learners literally move evidence until a pattern emerges. The gallery walk adds a peer-review layer (SEP 7: Engaging in Argument from Evidence) by requiring groups to evaluate each other's arrangements and pose questions, mirroring how scientists scrutinise each other's data displays.	Collect the sticky notes from the gallery walk. Scan for: (1) groups that correctly identified that elements with the same outer-shell electron count form a vertical column (group) — these groups understand the core concept; (2) groups whose sticky-note question reveals confusion between 'number of shells' (period) and 'outer-shell electrons' (group) — use these questions to frame the whole-class discussion in Phase 3. Observable indicator of success: every group can orally state 'Elements in the same column have the same number of outer-shell electrons' before Phase 3 begins.

	observation and one question on a sticky note left on that desk.	yours does not?' After the gallery walk, cold-call 2 groups to share the pattern they found. Ask the class: 'Raise your hand if your prediction from Phase 1 matches what you found.' Count raised hands aloud and record the number on the board.		
Explain Phase  VIDEO: Quantum numbers Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Electron Cloud Atomic Model (Flexbook) Source: CK-12  Search ARES for similar readings	Whole-class discussion anchored by the card-sort results. Learners are guided to: (a) define 'period' as a row corresponding to the number of occupied electron shells, and 'group' as a column corresponding to the number of outer-shell (valence) electrons; (b) state the periodic law in their own words — 'When elements are arranged in order of increasing atomic number, their properties repeat periodically'; (c) apply this to the fertiliser phenomenon: learners look at the CAN formula $\text{Ca}(\text{NO}_3)_2 \cdot \text{NH}_4\text{NO}_3$ and identify the group positions of Ca (Group 2, 2 outer electrons), N (Group 15/V, 5 outer electrons), and O (Group 16/VI, 6 outer electrons) using their annotated grid. They discuss in pairs for 3 minutes: 'Why does calcium form Ca^{2+} and not Ca^{3+} ? Use your card-sort evidence.' Pairs then share with the class. Learners annotate a blank 4×5 periodic table grid (elements 1–20 only) by placing each element card in its correct period and group position, then sketching the electron arrangement as concentric circles in the grid cell.	Open with: 'Every group found a repeating pattern. Scientists gave this pattern a name — the periodic law. Let's build the definition together from your own card-sort words.' Write learner-generated phrases on the board; then formalise: 'The periodic law states that when elements are arranged in order of increasing atomic number, their physical and chemical properties repeat periodically.' Ask the class: 'Which row does sodium sit in — and how do you know from the cards?' Allow 10 seconds WAIT TIME; cold-call a learner who had not yet contributed. After the grid annotation, pose the connecting question: 'Look at calcium on your grid. It has 2 outer-shell electrons. The CAN formula has Ca^{2+} . Our DAP formula has no calcium at all — it has ammonium, NH_4^+ . Where is nitrogen on your grid, and how many outer-shell electrons does it have?' Allow 15 seconds WAIT TIME; cold-call 2 learners. End this phase by asking learners to write one sentence completing: 'The periodic law explains the fertiliser formulae because _____. ' Collect these on exit slips.	Concept Definition Mapping (learner-generated → formalised). By building the definition of the periodic law from their own card-sort language before receiving the formal statement, learners construct meaning rather than receive it. The immediate application to the CAN/DAP phenomenon (connecting evidence to the driving question) uses the 'Anchoring Phenomenon Re-entry' strategy — every abstract concept is immediately re-grounded in the Kericho context, sustaining motivation and coherence across the 8-lesson sequence.	Collect the exit-slip sentences ('The periodic law explains the fertiliser formulae because...'). Sort into three piles: (A) sentences that correctly link valence electrons → ion charge → formula ratio; (B) sentences that reference the pattern but do not connect to the formula; (C) sentences that show no connection. Use pile-C learners as the target group for opening re-teaching at the start of Lesson 3. Observable success indicator: at least 70% of exit slips fall in pile A or B.
Driving Question Board (DQB) Creation  VIDEO:	Each learner receives two sticky notes of different colours. On the GREEN note they write one thing they can now explain about the fertiliser phenomenon that they	Say: 'Our Driving Question asks why CAN and DAP have different formulae. Two lessons in, what can we now say?' Give 2 minutes for silent sticky-note writing. As	Driving Question Board Protocol (DQB). The two-colour sticky-note system makes learner thinking visible and creates a cumulative public record of the class's growing	Count the ratio of green to orange notes. A healthy ratio is roughly 1:1 — if far more orange than green notes appear, learners have not consolidated enough

Quantum numbers Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Electron Cloud Atomic Model (Flexbook) Source: CK-12 Search ARES for similar readings	could not explain at the start of the lesson (e.g., 'I can now explain why calcium is in Group 2 and forms Ca^{2+}'). On the ORANGE note they write one new question the card sort has raised (e.g., 'Why does nitrogen appear in both CAN and DAP if it has 5 outer electrons — why doesn't it always form the same ion?'). Learners post both notes on the class Driving Question Board under the headers 'We Now Know' and 'We Still Wonder.' The teacher reads 3 orange notes aloud and asks the class: 'Is this question answered by what we know today, or do we need more lessons to answer it?' Learners vote by thumbs up/down.	learners post notes, read aloud 3 orange (wonder) notes — choose one that previews Lesson 3 content (ion formation and valency). Say: 'This question — [read the note] — is exactly what we will investigate in Lesson 3. Hold that thought.' Point to the prediction statements recorded at the start of Phase 1: 'Look at our predictions. [Learner name], your prediction said [quote it]. Was it supported by the card sort?' Allow 10 seconds WAIT TIME. Photograph the DQB for the ARES platform record.	understanding. The teacher's deliberate selection of orange notes that preview Lesson 3 content uses 'productive confusion' — surfacing an unresolved question to motivate the next investigation. This also reinforces the NGSS practice of asking questions and defining problems (SEP 1) as an ongoing scientific habit, not a one-off activity.	understanding from Phases 2–3 and the teacher should plan a brief re-teaching opener for Lesson 3. If almost no orange notes appear, learners may not yet be thinking deeply enough about the phenomenon — probe with targeted questions at the start of Lesson 3.
Model Building Phase  VIDEO: Quantum numbers Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Electron Cloud Atomic Model (Flexbook) Source: CK-12 Search ARES for similar readings	Learners return to the atomic model diagrams they began in Lesson 1 (concentric-circle Bohr models of hydrogen, lithium, sodium, and potassium). They now add a new layer of annotation: (a) label the outermost shell 'valence shell' and write the valence electron count; (b) draw a dotted boundary around the valence shell and label it 'the shell that determines group number'; (c) add a new element to their model set — they choose either calcium (Group 2, relevant to CAN) or phosphorus (Group 15, relevant to DAP) and draw its full Bohr model. In the final 3 minutes, learners write a 'Model Revision Note' at the bottom of their model page: 'My model now shows _____. The next thing my model cannot yet explain is _____. This prepares the model for Lesson 3's focus on ion formation.	Say: 'In Lesson 1 you drew atomic models. Today's evidence lets you make those models more powerful. Add the valence-shell label now.' Circulate and look specifically for: learners who draw calcium with 3 shells and 2 outer electrons (correct) vs. those who place 8 electrons in the third shell (incorrect — Ca has 2,8,8,2). For any learner drawing Ca incorrectly, ask: 'Your card from the sort — how many electrons total does calcium have? Walk me through filling the shells: first shell holds a maximum of...?' Allow 10 seconds WAIT TIME at each prompt. In the last 2 minutes, cold-call one learner to read their 'Model Revision Note' aloud. Ask the class: 'Does anyone's model revision note mention something the rest of us missed?' Take 2 responses. Close with: 'Next lesson, we will watch what happens when calcium gives away its 2 valence electrons — and that is the moment we will finally start	Cumulative Model Revision (NGSS SEP 2: Developing and Using Models). Each lesson in the sequence adds one layer of explanatory power to the learner's personal atomic model. The 'Model Revision Note' (what I can now explain / what I still cannot explain) externalises metacognition and directly connects the model to the driving question, ensuring that model building is purposeful rather than decorative. Choosing calcium or phosphorus as the new element to model deliberately connects model work to the fertiliser phenomenon.	Collect or photograph learners' model pages before the end of class. Check for: (1) correct Bohr model of calcium (2,8,8,2) with valence shell labelled — this is the key success indicator for this phase; (2) Model Revision Notes that name ion formation or bonding as 'the next thing my model cannot explain' — these learners are conceptually ready for Lesson 3; (3) Model Revision Notes that are blank or generic ('I don't know') — these learners need a bridging question at the start of Lesson 3 to re-engage.

		to see why Ca appears in CAN as Ca^{2+} .		
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D. TEACHER REFLECTION

After the lesson, consider the following questions to guide your preparation for Lesson 3:

1. CARD-SORT EVIDENCE: Did all groups independently discover the repeating outer-shell pattern, or did most groups need a teacher prompt? If more than half of your groups needed prompting, consider adding a partially completed reference table (showing elements 1–10 already sorted) to the Lesson 3 card-sort scaffolding for lower-readiness groups.
2. EXIT SLIPS: What proportion of exit slips fell in pile A (valence electrons → ion charge → formula ratio)? If fewer than 50% made this connection, open Lesson 3 with a 5-minute re-teaching segment using the calcium example: 'Calcium has 2 outer electrons — when it loses them, what charge does it carry, and what does that tell us about the formula it forms?'
3. DQB ORANGE NOTES: Were the unanswered questions mostly about ion formation (good — that is Lesson 3) or about topics far ahead in the sequence (e.g., bonding, rates of reaction)? If learners are jumping far ahead, use the DQB update at the start of Lesson 3 to reorient them: 'We will get there — but our next step is ion formation, because without that, the formula makes no sense.'
4. EQUITY CHECK: Did female learners and learners from tea-farming families in Kericho contribute proportionally to the card-sort discussion and cold-call responses? The fertiliser phenomenon is specifically grounded in their home context — if these learners were quiet, consider restructuring cold-call targeting in Lesson 3 to explicitly invite their lived experience ('You may have seen these fertiliser bags at home — what do you notice about the label?').
5. MODEL QUALITY: Were calcium Bohr models mostly correct (2,8,8,2)? A common error is 2,8,10 — if you saw this frequently, begin Lesson 3 with a 3-minute shell-filling rule revision before moving to ion formation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners sorted 20 element cards by electron arrangement data and discovered that the number of outer-shell (valence) electrons repeats in a predictable pattern as atomic number increases — elements with 1 outer electron appear at regular intervals, as do elements with 2, 3, and so on up to 8 (noble gases).
What did I learn?	The repeating pattern of valence electrons is called the periodic law. Elements are arranged in the periodic table in rows called periods (which correspond to the number of occupied electron shells) and columns called groups (which correspond to the number of valence electrons). Calcium (in CAN) is in Group 2 because it has 2 valence electrons; nitrogen (in both CAN and DAP) is in Group 15 because it has 5 valence electrons.
How does this explain the phenomenon?	The periodic law begins to explain why CAN and DAP have different formulae: the number of valence electrons determines how many electrons an atom can lose or gain to become stable, which in turn determines the charge of the ion it forms, which determines the ratio in which elements combine in a compound. Calcium's 2 valence electrons mean it forms Ca^{2+} — a fact that will directly explain the '2' in $\text{Ca}(\text{NO}_3)_2$ once we study ion formation in Lesson 3.

LESSON 3 (80 minutes): Stability, Electron Affinity, and the Noble Gas Rule

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate why noble gases are unreactive and how other atoms achieve stability by losing or gaining electrons, establishing the conceptual foundation for ion formation and fixed combining ratios in compounds.
Knowledge	Learners examine electron arrangement of atoms of an element and explain the stability and electron affinity of atoms of elements, and describe ion formation for the first 20 elements of the periodic table.
Skills	Learners draw dot (.) and cross (x) diagrams to illustrate ion formation; discuss with peers how atoms of elements acquire stability (losing or gaining electrons) relating to the atom's ability to attain a positive or negative charge (cations and anions); discuss the unreactive nature of the noble gases.
Attitudes	Learners appreciate the importance of the structure of the atom in the development of the periodic table and recognise and respect the input of every group member as they discuss how to determine valence and oxidation numbers of elements and radicals.
Key Inquiry Question	Why is electron arrangement significant?
Purpose in Storyline	In Lesson 2 learners mapped the electron arrangements of the first 20 elements and noticed the repeating pattern of outer-shell electrons. In this lesson they explain WHY that pattern matters — noble gases are stable because their outer shells are full, and every other element 'wants' the same stability. This directly answers why calcium always loses 2 electrons to become Ca^{2+} and nitrogen always gains 3 to become N^{3-} , the very ions that appear in the CAN and DAP fertiliser formulae on the Kericho tea farm.
Safety Notes	No hazardous chemicals are used in this lesson. If printed fertiliser bag labels or physical samples of CAN/DAP crystals are brought to class, learners must not taste the crystals and should wash hands after handling. Digital device use should follow school cyber-safety guidelines.

B. LESSON OVERVIEW

Learners begin by predicting which elements they think are the 'least willing' to react and why. They then observe evidence from three sources — a short animated simulation of noble-gas electron shells, a structured card-sorting activity using electron-arrangement data from Lesson 2, and a guided dot-and-cross diagram construction exercise — to build the concept that a full outer shell confers stability. In the Explain phase learners formalise the ideas of electron affinity, cation formation (metals losing electrons) and anion formation (non-metals gaining electrons), using calcium and nitrogen as anchor cases tied directly to the CAN and DAP fertiliser phenomenon. The DQB is updated to record the new mechanism, and learners revise their atomic model to show ion charges.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Kinetic molecular theory of gases	Learners are shown two printed fertiliser bag labels (CAN and DAP) — the same bags from the phenomenon introduction — and	Display the two fertiliser bag labels prominently. Write Ca^{2+} and N^{3-} on the board in large text and say: 'These two ions are hiding inside	Elicit-and-Record Predictions: by publicly committing predictions before instruction, learners activate prior knowledge from	Teacher scans prediction slips for two observable indicators: (1) Does the learner reference the outer shell electron count (2 for

Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Electron Cloud Atomic Model (Flexbook) Source: CK-12 Search ARES for similar readings	the teacher writes on the board: Ca^{2+} and N^{3-}. Working individually for 3 minutes, learners write responses to the prompt: 'Calcium is in Group 2 and has 2 outer electrons. Nitrogen is in Group 5 and has 5 outer electrons. Predict: how many electrons does each atom lose or gain to become the ion shown on the board? Why that number and not a different one?' Learners then share predictions with a shoulder partner for 2 minutes. The teacher cold-calls 4 pairs — two who predict a pattern and two whose predictions differ — and records all predictions verbatim on the 'Our Predictions' section of the class Driving Question Board without affirming or correcting any response.	the white crystals on the Kericho tea farm. Your job right now is NOT to get the right answer — it is to commit to a prediction so your brain has something to test.' Distribute the prediction slips (half-sheet paper). After 3 minutes of individual writing, say: 'Turn to your shoulder partner. You have 2 minutes — compare your numbers and your reasons.' [WAIT for full 2 minutes.] Then: 'I am going to cold-call four pairs. I need your honest prediction, not a corrected one.' Cold-call Pair 1 and Pair 2 (ask: 'What number did you predict for calcium, and what was your reasoning?'); cold-call Pair 3 and Pair 4 (ask: 'Did your partner agree with you? Where did you disagree?'). Record each prediction on the DQB under 'What we think we know.' Say: 'Every single prediction on this board will be tested in the next 40 minutes. Keep your slip — you will return to it.'	Lesson 2 (electron arrangements) and create cognitive dissonance that the Observe phase will resolve. Recording predictions on the DQB without evaluation signals that scientific thinking begins with a testable claim, not a correct answer.	Ca, 5 for N) in their reasoning? (2) Does the learner connect the ion charge to the number of electrons moved? Learners who write 'Ca loses 2 because it has 2 outer electrons' demonstrate readiness for the Explain phase. Learners who write random numbers or leave reasoning blank are flagged for targeted questioning during the Observe phase.
Observe Phase VIDEO: Kinetic molecular theory of gases Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Electron Cloud Atomic Model (Flexbook) Source: CK-12 Search ARES for similar readings	Learners engage with three sequential evidence sources. EVIDENCE SOURCE A (10 min): The teacher plays a 3-minute animation (or displays a teacher-drawn sequence on the board if no device is available) showing the electron shells of He (2), Ne (2,8), and Ar (2,8,8), pausing after each to allow learners to count outer electrons and record in their notebooks: element name, electron arrangement, number of outer electrons, and one observation about what they notice. EVIDENCE SOURCE B (12 min): In groups of 4, learners receive a set of 20 element cards (prepared from Lesson 2 data) each showing: element name,	EVIDENCE SOURCE A — Before playing the animation, say: 'As you watch, count the electrons in the OUTER shell only. Write the number in your notebook. Do not write anything else yet.' Play/draw the He shell, PAUSE for 10 seconds, then ask: 'Outer electrons for helium?' [Cold-call 2 learners.] Repeat for Ne and Ar. After all three, ask: 'What is the same about helium, neon, and argon?' [WAIT 12 seconds.] Cold-call 3 learners. Write responses on the board. EVIDENCE SOURCE B — Distribute card sets. Say: 'Group captain reads the instruction aloud.' After sorting (allow 8 min), stop groups and ask: 'Which sub-pile is larger — the 1-	Progressive Evidence Layering with Card Sorting: Evidence Source A establishes the empirical anchor (noble gases have full outer shells). Evidence Source B operationalises pattern-recognition by having learners physically sort all 20 elements, revealing that metals cluster on the low-outer-electron side and non-metals on the high side. Evidence Source C translates the abstract electron-count into a concrete, KICD-mandated representational practice (dot-and-cross diagrams), making the electron transfer visible. The three sources build on each other so that by the end of Observe, learners have the evidence needed to construct the	Observable indicators: (1) In Source A, can the learner correctly state outer electron counts for He, Ne, Ar without prompting? (2) In Source B, does the group correctly place all metals (Na, Mg, K, Ca, Li) in the 1-2-3 pile and all non-metals needing electrons (N, O, F, Cl) in the 5-6-7 pile? Misplacements of H or N signal confusion about valence. (3) In Source C, a correct Na^+ diagram shows: the original 3-shell structure, the outer electron removed, a net charge of +1 labelled, and the remaining 2,8 arrangement. A correct Cl^- diagram shows the added electron in the outer shell giving 2,8,8 and a net charge of -1. Teacher circulates and records names of

	symbol, atomic number, and electron arrangement. Groups are instructed: 'Sort your cards into two piles — elements whose outer shell is FULL (like the noble gases) and elements whose outer shell is NOT full. Then, for the NOT-full pile, sub-sort into: (i) elements with 1, 2, or 3 outer electrons and (ii) elements with 5, 6, or 7 outer electrons.' Groups record their sorted lists. EVIDENCE SOURCE C (10 min): Each learner independently draws dot-and-cross diagrams (using the KICD-mandated dot-and-cross notation) for: (a) sodium (2,8,1) losing 1 electron to become Na^+ and (b) chlorine (2,8,7) gaining 1 electron to become Cl^-. Teacher circulates and stamps or ticks correct diagrams immediately.	2-3 outer electron pile, or the 5-6-7 pile?' Cold-call 2 group captains. Then ask: 'What do you notice about the elements in the 1-2-3 pile compared to the ones we identified as metals in Lesson 1?' [WAIT 10 seconds.] Cold-call 2 learners. EVIDENCE SOURCE C — Say: 'This is individual work. Draw the full shell diagram for sodium — all three shells. Then show one electron leaving. Then draw what sodium looks like after. Label the charge.' Circulate; if a learner draws the electron staying inside the atom, crouch and quietly ask: 'Where is that electron going? Is sodium pulling it back, or releasing it?' After 10 minutes, display a correct Na^+ diagram on the board and a correct Cl^- diagram and ask: 'Who found something different from what you see here? Show me your diagram.' Cold-call 1 learner who had an error and lead a whole-class correction without embarrassment: 'This is a common place to make a mistake — let's fix it together.'	'full-outer-shell = stability' rule themselves.	learners who cannot yet draw the post-transfer shell structure — these learners receive a targeted re-teaching prompt in Phase 3.
Explain Phase  VIDEO: Kinetic molecular theory of gases Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Electron Cloud Atomic Model (Flexbook) Source: CK-12  Search ARES for similar readings	The teacher leads a structured whole-class discussion to formalise three connected concepts: (1) The Noble Gas Rule — atoms are most stable when their outer shell is full (like noble gases: $\text{He}=2$, $\text{Ne}=2,8$, $\text{Ar}=2,8,8$); (2) Electron Affinity — the tendency of an atom to attract and hold electrons; non-metals (high electron affinity) gain electrons and become anions; metals (low electron affinity relative to non-metals) lose electrons and become cations; (3) Ion Formation for the Phenomenon — learners apply the rule to calcium ($2,8,8,2 \rightarrow$ loses 2 $\rightarrow \text{Ca}^{2+}$, achieving 2,8,8 like	Open the Explain phase by pointing to the DQB predictions and saying: 'Let us now use the evidence to build the explanation together. I am not going to tell you the answer — I am going to ask you questions until we build it as a class.' Ask: 'From Evidence Source A, what did He, Ne, and Ar all have in common?' [WAIT 10 seconds. Cold-call 2 learners.] Write on the board: FULL OUTER SHELL. Ask: 'From Evidence Source B, the metals were on which side of our sort?' [Cold-call 1 learner.] Write: METALS — few outer electrons. Ask: 'If a sodium atom loses its 1 outer electron,	Socratic Questioning with Anchored Explanation: the teacher uses a chain of questions that each require learners to apply evidence from the Observe phase rather than recall a definition. By ending each question sequence with a connection to the phenomenon (Ca^{2+} in CAN, N^{3-} in DAP), the explanation is not abstract chemistry — it is the mechanism behind the specific ions printed on the Kericho fertiliser bags. The prediction-slip annotation at the end of the phase operationalises metacognition: learners explicitly track how their thinking changed, a core CBE	Observable indicators: (1) The learner's written sentence about Ca^{2+} must reference (a) the original electron arrangement 2,8,8,2, (b) the loss of 2 electrons, and (c) achieving the noble gas configuration of argon (2,8,8). Sentences that only say 'calcium loses 2 electrons because it is in Group 2' without referencing noble gas stability are incomplete — teacher provides a written prompt: 'What does the resulting shell look like?'. (2) The dot-and-cross diagram for Ca^{2+} must show: 3 shells before (2,8,8,2), 2 shells after (2,8,8), and the charge labelled as 2+. The diagram for

	argon) and nitrogen (2,5 → gains 3 → N^{3-}, achieving 2,8 like neon). Each learner draws dot-and-cross diagrams for Ca^{2+} and N^{3-} in their notebooks and writes one sentence connecting the ion charge back to the fertiliser formula. The class then revisits prediction slips and each learner annotates their original prediction: correct, partially correct, or revised — with an explanation of what evidence changed their thinking.	what does its new outer shell look like?' [WAIT 12 seconds. Cold-call 2 learners.] Write: $Na^+ = 2,8 =$ same as Neon. Say: 'This is called achieving noble gas configuration. It is the driving force behind ion formation.' Introduce the term ELECTRON AFFINITY by writing it on the board: 'Electron affinity is the attraction an atom has for electrons. Chlorine has HIGH electron affinity — it pulls electrons toward itself. Sodium has LOW electron affinity relative to chlorine — it releases electrons more easily.' Now anchor to the phenomenon: 'Calcium is in Group 2. It has electron arrangement 2,8,8,2. How many electrons must it lose to achieve 2,8,8?' [WAIT 10 seconds. Cold-call 3 learners.] Write: $Ca \rightarrow Ca^{2+} + 2e^-$. Ask: 'Nitrogen has arrangement 2,5. It needs 8 in the outer shell to match neon. How many does it need to gain?' [WAIT 10 seconds. Cold-call 3 learners.] Write: $N + 3e^- \rightarrow N^{3-}$. Say: 'Now write one sentence in your notebook that answers this question: Why does calcium always form a 2+ ion and never a 3+ or 1+ ion?' Allow 3 minutes of silent writing. Cold-call 2 learners to read their sentences aloud. Correct misconceptions using: 'Almost — what would calcium's shell look like if it lost 3 electrons instead of 2? Would it match any noble gas?' Close by saying: 'Return to your prediction slip. Annotate it — what did you get right? What did the evidence change for you?'	learning-to-learn competency.	N^{3-} must show: 2 shells before (2,5), 2 shells after (2,8), and charge labelled as 3-. (3) Prediction-slip annotations reveal whether learners can identify which piece of evidence (noble gas pattern, card sort, dot-and-cross) changed their thinking — this metacognitive indicator is noted by the teacher for the teacher reflection.
Driving Question Board (DQB) Creation	Each group of 4 receives three sticky notes in different colours: GREEN for 'What we now understand', ORANGE for 'What	Distribute sticky notes by colour. Say: 'GREEN note — write ONE thing that your group can now explain that you could NOT explain	Colour-Coded DQB Contribution: the three-colour system separates understanding from uncertainty from curiosity, making the	Observable indicator: GREEN notes that contain the words 'noble gas', 'full outer shell', 'Ca^{2+}', or 'N^{3-}' in a coherent explanatory

 VIDEO: Kinetic molecular theory of gases Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Electron Cloud Atomic Model (Flexbook) Source: CK-12  Search ARES for similar readings	we are still unsure about', and BLUE for 'New questions this lesson created'. Groups spend 5 minutes writing one contribution per colour. The teacher facilitates a whole-class gallery walk (3 minutes) where learners read other groups' sticky notes silently. The teacher then reads aloud 2 selected ORANGE and 2 selected BLUE notes and explicitly connects them to upcoming lessons: 'This question about HOW calcium and nitrogen actually come together to form a compound — that is exactly what Lesson 4 will address when we study valency and chemical formulae.' The teacher also points to the original driving question written on the DQB — 'Why do the fertiliser bags carry such specific and different chemical formulae?' — and asks: 'Can someone now give me one more piece of the answer to our driving question than you could at the start of today?' Cold-calls 2 learners.	at the start of Lesson 3. Be specific: use the words calcium, nitrogen, electron, or noble gas.' [Allow 2 minutes.] 'ORANGE note — write ONE idea your group is still fuzzy about. Be honest — scientists are always partly uncertain.' [Allow 1.5 minutes.] 'BLUE note — write ONE brand-new question that today's lesson created for you. The best questions are ones we cannot yet answer.' [Allow 1.5 minutes.] Post all notes on the DQB in the designated columns. Lead the gallery walk: 'Walk silently — no talking — and read as many notes as you can in 3 minutes.' After the walk, read 2 ORANGE notes aloud and say: 'I am not going to answer these today — but I want you to notice that these questions are EXACTLY the ones that drove scientists like Dmitri Mendeleev to keep refining the periodic table.' Read 2 BLUE notes and connect to the lesson sequence. Finally point to the driving question and cold-call 2 learners for their updated partial answers.	epistemic state of the class visible to both teacher and learners. The gallery walk builds on the KICD Communication and Collaboration competency and ensures that quieter learners contribute ideas in writing before they are discussed orally. Explicitly linking ORANGE and BLUE notes to upcoming lessons models scientific practice — questions drive inquiry forward.	sentence demonstrate conceptual consolidation. GREEN notes that only restate facts ('calcium loses 2 electrons') without connecting to stability indicate surface-level recall. The teacher photographs the DQB at the end of the lesson for use in Lesson 4 planning. ORANGE notes citing confusion about 'why not lose 4 electrons instead of 2' reveal a misconception about stability that must be pre-addressed in Lesson 4's opening.
Model Building Phase  VIDEO: Kinetic molecular theory of gases Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Electron Cloud Atomic Model (Flexbook) Source: CK-12  Search ARES	Learners return to the cumulative atomic model they began building in Lesson 1 (a hand-drawn or digitally constructed diagram of the atom showing protons, neutrons, and electron shells). In Lesson 2 they annotated the model with the electron arrangement of the first 20 elements. Today each learner adds a new layer to their model: (a) they draw BEFORE and AFTER shell diagrams for calcium ($\text{Ca} \rightarrow \text{Ca}^{2+}$) and nitrogen ($\text{N} \rightarrow \text{N}^{3-}$) directly onto or beside their cumulative model page; (b) they add a legend box with three terms and definitions in their own words:	Say: 'Open your cumulative atomic model — the one you have been building since Lesson 1. Today it grows again.' Display the teacher's own model under a projector or on the board as a reference (not a copy template). Say: 'Add your before-and-after shell diagrams for calcium and nitrogen. This is the most important thing you will draw today — these are the actual ions inside the fertiliser bag in Kericho.' Allow 7 minutes of individual model work. Circulate and narrate observations aloud to the class: 'I can see Akinyi has drawn three shells for calcium before the	Cumulative Model Revision: the model-building phase operationalises the NGSS Science and Engineering Practice of Developing and Using Models. By requiring learners to physically add new content to an existing model (rather than starting fresh), the lesson makes conceptual progression visible and tangible. The phenomenon-linking sentence at the bottom of the model page ensures that every representational act connects to the real-world anchor, preventing the diagrams from becoming isolated drill exercises. Peer	Observable indicators for model quality: (1) The Ca before-diagram correctly shows 4 shells (2,8,8,2); the Ca^{2+} after-diagram correctly shows 3 shells (2,8,8) labelled 2+. (2) The N before-diagram correctly shows 2 shells (2,5); the N^{3-} after-diagram correctly shows 2 shells (2,8) labelled 3-. (3) The legend box contains student-generated (not copied) definitions. (4) The phenomenon sentence mentions both ions and connects each to a named noble gas (argon for Ca^{2+}, neon for N^{3-}). Teacher collects 5 models at random at the end of class for written feedback to be

for similar readings	NOBLE GAS RULE, ELECTRON AFFINITY, CATION / ANION; (c) they write one sentence at the bottom of their model page that explicitly connects the ion charges back to the phenomenon: 'Ca²⁺ forms in CAN fertiliser because calcium achieves argon's noble gas configuration by losing 2 electrons; N³⁻ forms in both CAN and DAP fertilisers because nitrogen achieves neon's noble gas configuration by gaining 3 electrons.' Learners share their revised model with their shoulder partner for a 2-minute peer check using one prompt: 'Does your partner's model show the electron count before AND after, with the correct charge labelled?'	transfer and two shells after — excellent. I can see James is drawing four shells — James, how many shells does calcium have? Count from the electron arrangement 2,8,8,2.' After 7 minutes: 'Add your legend box — three terms, YOUR own words, not copied from the board.' [Allow 3 minutes.] 'Now write your one connecting sentence about the phenomenon. Use the words CAN, DAP, Ca²⁺, N³⁻, noble gas, and configuration.' [Allow 2 minutes.] 'Share with your partner. Use the peer-check prompt on the board: Does your partner's model show the electron count before AND after, with the correct charge labelled? Give your partner one specific piece of praise and one specific suggestion.' [Allow 2 minutes.] Close by saying: 'Your model now has three lessons of evidence built into it. In Lesson 4 we will use these ions and their charges to figure out why calcium and nitrogen combine in the exact ratio that gives us Ca(NO₃)₂ — the formula on the CAN bag.'	checking using a specific structural prompt (before/after counts, charge label) turns partner work into genuine formative feedback rather than social chat.	returned at the start of Lesson 4.
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record brief notes for Lesson 4 planning: (1) PREDICTION QUALITY — Did learners' predictions reference outer-shell electron counts, or were they guessing? If fewer than half connected the ion charge to the outer electron count in their prediction slips, Lesson 4 should open with a brief retrieval practice on electron arrangements before moving to valency. (2) CARD-SORT ACCURACY — Did any groups place hydrogen (1 outer electron) or nitrogen (5 outer electrons) in the wrong sub-pile? Confusion about hydrogen's placement (it can lose OR gain 1 electron) is a productive misconception to address explicitly in Lesson 4 when writing chemical formulae. (3) DOT-AND-CROSS FLUENCY — Were learners able to draw the post-transfer shell diagram for Ca²⁺ (a 3-shell → 2-shell transformation) without scaffolding? If more than 30% of learners drew 4 shells for Ca²⁺ after the transfer, allocate the first 5 minutes of Lesson 4 to a corrective worked example. (4) DQB ORANGE NOTES — Were there notes expressing confusion about why atoms stop at the noble gas configuration and do not keep losing electrons? This misconception (e.g. 'why doesn't Ca lose all 20 electrons?') indicates learners do not yet have a feel for the energy cost of removing core electrons — a qualitative explanation using the concept of increasing nuclear attraction on inner-shell electrons is sufficient at this level and can be addressed in a 2-minute teacher explanation at the start of Lesson 4. (5) PHENOMENON CONNECTION — In the model-building phase, did learners' connecting sentences correctly name argon as the noble gas target for Ca²⁺ and neon for N³⁻? If fewer than 60% made this specific noble-gas-name connection, the Lesson 4 warm-up should revisit the noble gas configurations of the first 20 elements using the electron-arrangement cards from Lesson 2.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed three pieces of evidence: (1) noble gases (He, Ne, Ar) all have full outer electron shells; (2) when the first 20 elements are sorted by outer electron count, metals cluster on the low side (1–3) and reactive non-metals on the high side (5–7); (3) dot-and-cross diagrams show that Na loses 1 electron to achieve neon's configuration (Na^+) and Cl gains 1 electron to achieve argon's configuration (Cl^-).
What did I learn?	Atoms are most stable when their outer shell is full — matching a noble gas configuration. Atoms with few outer electrons (metals) lose electrons and form positively charged cations. Atoms with nearly full outer shells (non-metals) gain electrons and form negatively charged anions. Calcium (2,8,8,2) loses 2 electrons to achieve argon's configuration (2,8,8), forming Ca^{2+} . Nitrogen (2,5) gains 3 electrons to achieve neon's configuration (2,8), forming N^{3-} . Electron affinity describes an atom's tendency to attract electrons.
How does this explain the phenomenon?	The specific ion charges of Ca^{2+} and N^{3-} are not arbitrary — they are determined by how many electrons each atom must lose or gain to achieve the nearest noble gas electron configuration. This is why calcium always forms a 2+ ion (never 1+ or 3+) and nitrogen always forms a 3– ion. These fixed charges will directly determine the combining ratio of calcium and nitrogen ions in compounds, explaining why the CAN fertiliser formula is $\text{Ca}(\text{NO}_3)_2$ and not some other ratio.

LESSON 4 (80 minutes (2 × 40-minute lessons)): Ion Formation: Dot-and-Cross Diagrams

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners draw dot-and-cross diagrams to illustrate ion formation for selected elements from the first 20 (Na^+ , Mg^{2+} , Al^{3+} , Cl^- , O^{2-} , Ca^{2+}) and connect ion charges directly to periodic table group numbers, building towards explaining why CAN and DAP carry different formulae.
Knowledge	Learners describe ion formation for the first 20 elements of the periodic table, explaining how atoms of elements acquire stability by losing or gaining electrons to attain a positive or negative charge (cations and anions), and how the resulting ionic charge relates to the element's group in the periodic table.
Skills	Learners draw dot (•) and cross (×) diagrams to illustrate ion formation; observe and interpret animations/simulations on ion formation using available digital devices; present and justify diagrams to peers; and use group-number patterns to predict the charge an ion will carry.
Attitudes	Learners appreciate the importance of the structure of the atom in the development of the periodic table, and take turns playing roles when drawing dot-and-cross diagrams to illustrate ion formation, demonstrating unity and respect for every group member's input.
Key Inquiry Question	Why is electron arrangement significant? / How are elements arranged in the periodic table?
Purpose in Storyline	In Lessons 1–3 learners established atomic structure, electron configuration, and how the periodic table is organised by electron arrangement. In Lesson 4 they now see what happens when atoms of elements such as calcium (in CAN) or nitrogen/phosphorus (in DAP) gain or lose electrons to become ions — the very charged particles that combine in fixed ratios to produce the specific formulae on those fertiliser bags. This lesson is the crucial bridge between 'electron arrangement' and 'chemical formula'.
Safety Notes	No hazardous chemicals are handled in this lesson. Learners use pencils, paper, and digital devices only. Ensure digital devices are handled responsibly and screen time is managed. If printed animations are used, keep the classroom tidy to avoid tripping hazards.

B. LESSON OVERVIEW

Learners open by revisiting their Lesson 3 electron-configuration cards for Na, Mg, Al, Cl, O, and Ca — six elements that appear directly in the CAN and DAP fertiliser formulae. Through a structured predict–observe–explain sequence they first predict which electrons will be lost or gained, then watch a short animation/simulation of ion formation on available digital devices, and finally draw dot-and-cross diagrams in groups. Each group presents one ion diagram and ties the ionic charge back to the element's group number. The lesson closes with a DQB update and a cumulative model revision that adds the concept of ionic charge to learners' growing atomic model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Kinetic molecular theory of gases Source: Khan	Learners work individually (3 minutes silent think-time) then in pairs (2 minutes) to answer the prompt card: 'Calcium is in Group 2 of the periodic table and has the	Display the two fertiliser bag labels (CAN and DAP) on the board or projector alongside a blank periodic table showing only group numbers. Say: 'Last lesson we	Predict-before-observe (NGSS Science & Engineering Practice 2 — Developing and Using Models): Requiring a written, reasoned prediction activates prior	Observable indicator: Every learner has a written prediction with a stated reason in their exercise book before Phase 2 begins. Teacher scans books

Academy (English - US curriculum) Search ARES for similar videos HTML: Transition Metal Ion Formation (Flexbook) Source: CK-12 Search ARES for similar readings	electron arrangement 2,8,8,2. Nitrogen is in Group 5 and has the arrangement 2,5. Predict: (a) which electrons each atom is most likely to lose or gain, and (b) what charge the resulting ion will carry. Write your prediction in your exercise book and give a reason.' Pairs then share with the quad (group of 4) before a whole-class cold-call share-out. Learners also re-examine the CAN bag label — $\text{Ca}(\text{NO}_3)_2 \cdot \text{NH}_4\text{NO}_3$ — and circle the Ca, N, and O symbols, predicting in writing: 'Why do you think calcium appears as Ca^{2+} in this compound, not Ca or Ca^{3+}?'	mapped how electrons are arranged in shells. Today we ask: what happens when an atom GIVES AWAY or TAKES electrons? Let us make a prediction before we see any evidence.' Distribute the prompt cards. Allow 3 minutes of individual silent thinking — circulate and WAIT; do not prompt yet. After 3 minutes say: 'Share your prediction with your partner — you have 2 minutes.' After pair discussion, cold-call 3 pairs (select one high-confidence, one uncertain, one who has written a different answer): 'Akinyi, what did your pair predict for calcium?' → WAIT 10 seconds. 'Ouma, do you agree or disagree — and why?' → WAIT 12 seconds. 'Fatuma, your pair had a different prediction — tell us.' Record all three predictions on the board under the heading PREDICTED ION CHARGES — do NOT confirm or deny. Anchor the predictions to the phenomenon: 'Hold these predictions. By the end of today we will know whether the formula $\text{Ca}(\text{NO}_3)_2$ makes sense with what your atoms actually do.'	knowledge of electron configuration and forces learners to surface their mental model of atomic stability before encountering the correct answer. Recording competing predictions on the board creates cognitive dissonance that motivates careful observation in Phase 2.	during circulation and notes (a) whether learners correctly identify the outer-shell electrons as the relevant electrons, and (b) whether predicted charges match group number. Mismatches between predicted charges and group numbers flag misconceptions about which shell electrons are involved — teacher notes these learners for targeted questioning in Phase 3.
Observe Phase VIDEO: Kinetic molecular theory of gases Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Transition Metal Ion Formation (Flexbook) Source: CK-12	Activity A — Animation/Simulation (15 minutes): In groups of 4, learners open the ARES offline simulation 'Ion Formation — First 20 Elements' on available digital devices (or teacher projects centrally if devices are limited). Each group is assigned two ions: Group 1 → Na^+ and Cl^-; Group 2 → Mg^{2+} and O^{2-}; Group 3 → Al^{3+} and Ca^{2+}; Group 4 → K^+ and N^{3-}; any additional groups rotate the same assignments. Learners watch the animation twice: first time for general understanding,	Before starting the simulation say: 'You are scientists gathering evidence. Your job is NOT to copy — it is to OBSERVE and record in the table. Watch for: which shell loses or gains electrons, and what the atom looks like before and after.' Pause the simulation after the sodium example and ask the whole class: 'What did you notice about which shell the electron left from?' Cold-call 2 learners → WAIT 10 seconds each. Resume simulation. During Activity B, circulate to each group. At Group 1	Multimodal evidence gathering + model construction (NGSS SEP 2 and SEP 4 — Analysing and Interpreting Data; Developing and Using Models): Learners first gather evidence from the animation (dynamic, visual), then encode that evidence into a static representational model (dot-and-cross diagram). The two-pass animation strategy (first for overview, second for structured recording) prevents passive watching and ensures every learner generates a personal	Observable indicators: (1) Every learner's observation table has all five columns completed for both assigned ions before diagram drawing begins. (2) Dot-and-cross diagrams show square brackets, correct electron counts (verified against electron configuration from Lesson 3), correct charge notation outside brackets, and a distinction between dots and crosses. Teacher collects 2–3 individual books from each group to check electron counts against the periodic table. Common errors to

Search ARES for similar readings	second time completing an observation record table with columns: Element Group No. Electron arrangement Electrons lost/gained Ion formed Charge on ion. Activity B — Dot-and-Cross Diagram Construction (15 minutes): Using the observation table as evidence, each learner individually draws dot-and-cross diagrams for their two assigned ions in their exercise book following the convention: electrons of the original atom shown as dots (•), electrons gained shown as crosses (x), square brackets drawn around the ion symbol, and charge written outside top-right. Groups then compare individual diagrams, agree on a consensus version, and reproduce it neatly on an A3 sheet for presentation.	ask: 'Show me on your diagram where the electron that left sodium went.' At Group 2: 'How many crosses did you draw inside oxygen's brackets — why that number?' At Group 3: 'You drew Al^{3+} — count the electrons in your diagram. Does that number match your observation table?' Use the specific phrase: 'Remember, your dot-and-cross diagram must TELL THE STORY of what the animation showed — it is a model of evidence, not a drawing exercise.' If devices are unavailable, use the printed storyboard handout (6-frame sequence per ion) available in the ARES offline resource pack. Ensure all groups have completed the observation table before moving to diagram construction.	evidence record. The A3 group consensus diagram creates a shared artefact that will be critiqued in Phase 3.	watch for: drawing too few electrons in the ion (forgetting to show the full electron count of the ion, not just the outer shell); omitting brackets; writing the charge inside the brackets.
Explain Phase VIDEO: Kinetic molecular theory of gases Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Transition Metal Ion Formation (Flexbook) Source: CK-12 Search ARES for similar readings	Gallery Presentation (10 minutes): Groups post their A3 dot-and-cross diagrams on the wall. Each group nominates one spokesperson who presents in 90 seconds: (a) the ion diagram, (b) how many electrons were lost or gained, (c) the resulting charge, and (d) which group in the periodic table the element belongs to — making the explicit claim: 'The charge on [element]'s ion equals its group number / (8 minus its group number).' Audience groups use a peer-feedback card with two prompts: 'One thing the diagram shows clearly...' and 'One question I still have about this diagram...' Sensemaking Discussion (10 minutes): Teacher leads a whole-class discussion anchored on the question: 'We now have evidence that Na loses 1 electron → Na^+ (Group 1), Mg loses 2 → Mg^{2+} (Group 2), Al	During gallery presentations, teacher stands to the side and resists correcting errors immediately. After each presentation ask the audience: 'Does this diagram match what YOU saw in the simulation? Raise your hand if you drew it differently.' If hands are raised: 'Kamau, come show us your version — what is different and why?' → WAIT 12 seconds. After all presentations, draw a summary table on the board collaboratively with learners: Group Element Electrons lost/gained Ion charge. Cold-call one learner per row to fill in the table → WAIT 10 seconds per call (target at least 6 different learners across the table). After the table is complete, ask the class: 'What rule could a Kericho tea farmer use to predict the charge on any ion just by looking at the periodic table group number?' → Think-Pair-	Claim-Evidence-Reasoning (CER) gallery critique (NGSS SEP 6 — Constructing Explanations): By requiring each presenting group to state an explicit claim ('the charge equals the group number') backed by their diagram as evidence, and by having audience groups generate questions, the lesson enacts disciplinary literacy in science. The whole-class pattern-finding discussion (filling the summary table) moves learners from individual ion cases to a generalisable rule — the hallmark of scientific sensemaking. Writing the 3-sentence explanation consolidates the rule in learners' own language before it is applied to the phenomenon.	Observable indicators: (1) Each group's presentation explicitly states the relationship between group number and ionic charge — not just the charge value. (2) Every learner's 3-sentence written explanation contains the words 'electron arrangement,' 'group number,' and 'ion charge' used correctly. (3) The summary table on the board is correctly completed with learner input (teacher only prompts, does not write). Teacher uses the peer-feedback cards collected after the gallery to identify lingering misconceptions — specifically any learner who writes that 'all atoms form positive ions' or who conflates mass number with ion charge.

	loses 3 → Al^{3+} (Group 3), O gains 2 → O^{2-} (Group 6), Cl gains 1 → Cl^- (Group 7). What pattern do you see between group number and ion charge? Learners write a 3-sentence explanation in their books linking electron arrangement → ion formation → ion charge → group number. Finally, teacher returns to the CAN bag: 'We have Ca^{2+} and NO_3^- . Now you know WHY calcium is Ca^{2+} — it is in Group 2. In our next lessons we will use these charges to explain why the formula is $\text{Ca}(\text{NO}_3)_2$ and not CaNO_3 .'	Share, 2 minutes. Cold-call 2 pairs. Write the agreed rule on the board: 'Metals in Groups 1, 2, 3 lose electrons equal to their group number → positive ions. Non-metals in Groups 6, 7 gain electrons to make up 8 → negative ions.' Explicitly anchor: 'This rule is why the formulae on those fertiliser bags are what they are — not random, but determined by electron structure.'		
Driving Question Board (DQB) Creation  VIDEO: Kinetic molecular theory of gases Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Transition Metal Ion Formation (Flexbook) Source: CK-12  Search ARES for similar readings	Learners revisit the class DQB (established in Lesson 1). Working individually for 3 minutes, each learner writes on a sticky note (or index card) a response to one of two prompts: PROMPT A — 'One thing I can now explain about the fertiliser formula that I could NOT explain before today.' PROMPT B — 'One question the ion diagrams raised for me that the DQB does not yet answer.' Learners post Prompt A notes under the column 'NOW I CAN EXPLAIN...' and Prompt B notes under 'STILL WONDERING...' The teacher reads 3–4 Prompt B notes aloud and the class votes (thumbs up) on which unanswered questions should be prioritised in Lessons 5–8. Selected questions are circled on the DQB.	Say: 'Our Driving Question asks why those two fertiliser bags have such specific and different formulae. You have just built a key piece of evidence. Take 3 minutes — think quietly — and write your sticky note.' Circulate silently during the 3 minutes. After posting, read 4 Prompt B questions aloud verbatim without editing: 'Juma writes: If Ca^{2+} has a charge of 2+, why does it need TWO nitrate ions — what charge does nitrate carry?' — 'Who else is wondering something like this? Thumbs up.' Tally. This surfaces the concept of valency and formula writing that Lesson 5 will address. Conclude: 'Your Prompt B questions are exactly the questions chemists asked — and the answers ARE in the periodic table. Our DQB is now guiding where we go next.'	Driving Question Board metacognitive update (NGSS SEP 1 — Asking Questions): The two-prompt structure separates consolidated knowledge (Prompt A) from productive uncertainty (Prompt B). Voting on unanswered questions gives learners agency over the inquiry trajectory and models how scientific communities prioritise research questions. The DQB physically accumulates across all 8 lessons, making the growth of understanding visible.	Observable indicators: (1) Every learner has posted at least one sticky note. (2) Prompt A notes correctly reference ion charge and/or group number — vague notes ('I learned about atoms') indicate a learner who has not consolidated the lesson objective and needs follow-up. (3) Prompt B notes that identify the charge-ratio problem (e.g., 'why Ca^{2+} needs two NO_3^- ') confirm readiness for Lesson 5 on valency and formula writing. Teacher photographs the DQB at the end of every lesson for the ARES platform record.
Model Building Phase  VIDEO: Kinetic molecular theory of gases Source: Khan	Learners retrieve the cumulative atomic model they began revising in Lesson 1 (a hand-drawn or digitally created diagram of an atom with shells). Today they add a new layer: on a fresh page they draw a BEFORE/AFTER model for	Say: 'Your model has grown since Lesson 1. Today you are adding the DYNAMIC part — what atoms DO with their electrons. Open your model booklet / exercise book to the model page.' Give 8 minutes for individual model drawing.	Cumulative model revision (NGSS SEP 2 — Developing and Using Models): Requiring the BEFORE/AFTER structure forces learners to make the transformation explicit rather than just labelling an ion in isolation.	Observable indicators: (1) Every learner's BEFORE panel shows the correct neutral electron arrangement matching the Lesson 3 reference chart. (2) The AFTER panel shows the correct electron count for the ion (e.g., Na^+ has 10

Academy (English - US curriculum) Search ARES for similar videos HTML: Transition Metal Ion Formation (Flexbook) Source: CK-12 Search ARES for similar readings	one cation and one anion of their choice from today's lesson. The BEFORE panel shows the neutral atom with electron shells labelled. The AFTER panel shows the ion with the correct electron count, square brackets, and charge. Below the two panels they write one sentence: 'This ion forms because [element] achieves a stable electron arrangement of [noble gas configuration] by [losing/gaining] [n] electron(s), and this is why its ion has a [n+/-] charge.' Groups share their models with one other group and check each other's electron counts against the Lesson 3 electron-configuration reference chart. Any corrections are made in a different colour pen so the revision process is visible.	Circulate and at each desk ask one targeted question: 'What noble gas configuration is your ion trying to reach?' → WAIT 10 seconds. 'How does that connect to the stable outer-shell number we discussed?' After peer-checking, bring the class together: 'Hold up one model that was CORRECTED in a different colour. Wanjiku — what did your partner correct and why?' → WAIT 12 seconds. 'This is exactly what scientists do — they revise their models when new evidence arrives. The fertiliser formula is not a mystery anymore — it is a RESULT of what these atoms do.' Close the lesson by pointing back to the CAN bag: 'Next lesson we use these ion charges to derive the formula $\text{Ca}(\text{NO}_3)_2$ step by step — valency and oxidation numbers.'	Writing the explanatory sentence in their own words consolidates the causal chain: electron arrangement → desire for stability → electron loss/gain → ionic charge. The different-colour peer-correction strategy makes scientific revision a visible, valued practice rather than an admission of error.	electrons, not 11), with square brackets and charge notation. (3) The written sentence contains a correctly named noble gas (e.g., 'achieves the configuration of neon'). Teacher collects model booklets from 5 randomly selected learners at the end of the lesson to assess depth of explanation before Lesson 5.
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record brief notes in your ARES teacher journal:

1. **DIAGRAM ACCURACY:** When I scanned the dot-and-cross diagrams during the gallery, how many groups correctly placed square brackets AND the charge notation AND the correct total electron count in the ion? If fewer than 3 out of 4 groups achieved all three, plan a 5-minute correction slot at the start of Lesson 5 using a worked example on the board (e.g., $\text{Mg} \rightarrow \text{Mg}^{2+}$).
2. **GROUP NUMBER RULE:** During the sensemaking discussion, did learners independently articulate the group-number-to-ion-charge rule, or did I have to supply it? If I supplied it, try the think-pair-share structure again in Lesson 5 with a new pattern (valency from group number) and allow longer wait time (15 seconds instead of 10).
3. **DQB QUALITY:** How many Prompt B sticky notes pointed towards the charge-ratio problem (why Ca^{2+} combines with 2 NO_3^-)? If fewer than half the class surfaced this, spend 3 minutes at the start of Lesson 5 revisiting the CAN formula with the explicit question: ' Ca^{2+} has a 2+ charge and NO_3^- has a 1- charge — what does that tell us about how many NO_3^- ions are needed to balance one Ca^{2+} '?
4. **EQUITY:** Did I cold-call learners from across the room (not only front rows)? Did I achieve gender balance across the 6+ cold-calls? Note which learners have not yet spoken publicly in this strand and target them in Lesson 5.
5. **PHENOMENON CONNECTION:** At the end of the lesson, could learners explain IN THEIR OWN WORDS why Ca^{2+} appears in CAN — not just recall it? If not, the model-building phase needs more time in future iterations.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners watched an animation/simulation of ion formation for elements from the first 20 of the periodic table (Na, Mg, Al, Cl, O,
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	Ca) and recorded in an observation table: electron arrangement before and after, number of electrons lost or gained, the ion formed, and the resulting charge.
What did I learn?	Atoms of metallic elements in Groups 1, 2, and 3 lose electrons equal to their group number to form positively charged ions (cations: Na^+ , Mg^{2+} , Al^{3+} , Ca^{2+}); atoms of non-metallic elements in Groups 6 and 7 gain electrons to complete their outer shell and form negatively charged ions (anions: O^{2-} , Cl^-). The ionic charge is determined by the atom's electron arrangement and its position in the periodic table — the same inner-atomic structure that organises the periodic table.
How does this explain the phenomenon?	The specific formulae on the CAN ($\text{Ca}(\text{NO}_3)_2 \cdot \text{NH}_4\text{NO}_3$) and DAP ($(\text{NH}_4)_2\text{HPO}_4$) fertiliser bags used on the Kericho tea farm are not arbitrary. Calcium, in Group 2, always forms Ca^{2+} by losing its two outer electrons to achieve a stable noble-gas configuration. Nitrogen and oxygen form ions whose charges are likewise fixed by their electron arrangements. These fixed ionic charges mean that elements must combine in specific ratios to produce electrically neutral compounds — which is why the formula is $\text{Ca}(\text{NO}_3)_2$ and not CaNO_3 . The 'hidden pattern' the learner noticed on the fertiliser bag is the direct consequence of ion formation driven by electron arrangement.

LESSON 5 (80 minutes (2 × 40-minute lessons)): Valency, Oxidation Numbers, and Radicals: Unlocking the Hidden Pattern in Fertiliser Formulae

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners derive valency from group position and electron arrangement, assign oxidation numbers to elements and radicals, and use these to write correct chemical formulae for compounds found in Kenyan fertilisers and everyday substances.
Knowledge	Valency is determined by the number of electrons an atom loses, gains, or shares to achieve a stable electron configuration; oxidation numbers represent the charge an atom carries in a compound or ion; radicals are groups of atoms that carry a fixed charge and behave as a unit in chemical reactions. The KICD-mandated radicals are: ammonium ion (NH_4^+), sulphate (SO_4^{2-}), nitrate (NO_3^-), hydroxide (OH^-), carbonate (CO_3^{2-}), phosphate (PO_4^{3-}), and hydrogen carbonate (HCO_3^-).
Skills	Deriving valency from group number and ion charge; assigning oxidation numbers to elements in compounds; applying the 'criss-cross' rule to write correct chemical formulae; identifying radicals by name, formula, and charge; constructing formulae for CAN [$\text{Ca}(\text{NO}_3)_2 \cdot \text{NH}_4\text{NO}_3$] and DAP [$(\text{NH}_4)_2\text{HPO}_4$] from first principles.
Attitudes	Appreciate that the specific chemical formulae on fertiliser bags are not arbitrary labels but precise records of valency relationships that determine how compounds behave in soil and in plants; respect the systematic nature of chemistry as a tool for solving agricultural and environmental problems in Kenya.
Key Inquiry Question	Why is electron arrangement significant? How are elements arranged in the periodic table?
Purpose in Storyline	In Lessons 1–4, learners established atomic structure, electron configuration, ion formation, and the logic of the periodic table. Lesson 5 now builds the critical bridge between electron arrangement and the written formula on a fertiliser bag. Learners discover that valency is simply a consequence of how many electrons an atom must lose, gain, or share — and that radicals like NO_3^- and PO_4^{3-} are the very polyatomic ions that appear in CAN and DAP. By the end of this lesson, the learner in Kericho can begin to decode why $\text{Ca}(\text{NO}_3)_2$ must have two nitrate groups and why $(\text{NH}_4)_2\text{HPO}_4$ must have two ammonium groups — answers that emerge directly from valency and oxidation number rules.
Safety Notes	No hazardous chemicals are used in this lesson. If fertiliser samples are handled for the predict phase, learners should wash hands after touching granules and must not taste any sample. Dust from granular fertilisers can irritate eyes — keep bags sealed and handle in a well-ventilated space.

B. LESSON OVERVIEW

Learners open by revisiting the two fertiliser bags (CAN and DAP) from the anchoring phenomenon and are challenged to predict why the formulae have specific subscript numbers. Through a guided derivation activity, they connect group position to valency for the first 20 elements. The teacher then introduces oxidation numbers as a formalisation of the same idea. Learners are introduced verbatim to the seven KICD-mandated radicals and their charges. Using the criss-cross rule, small groups construct formulae for a set of Kenyan compounds — including calcium nitrate, ammonium sulphate, and sodium carbonate — before deriving CAN and DAP from first principles. The lesson closes with DQB updates and a cumulative model revision linking valency to the phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Simplifying Radicals Review Principles - Basic Source: CK-12  Search ARES for similar videos  HTML: Observation: Penny Oxidation (PLIX) Source: CK-12  Search ARES for similar readings	Learners examine projected images of the CAN bag $[\text{Ca}(\text{NO}_3)_2 \cdot \text{NH}_4\text{NO}_3]$ and the DAP bag $[(\text{NH}_4)_2\text{HPO}_4]$ from Kericho. Each learner silently writes answers to three prompt questions in their science journal: (1) Why does calcium appear once but nitrate appears twice in CAN? (2) Why does ammonium appear twice but phosphate once in DAP? (3) What property of calcium, ammonium, or nitrate could force these specific numbers? Learners share predictions with a shoulder partner (Think-Pair-Share) before two or three pairs share with the class. Predictions are recorded on a dedicated 'Our Predictions' section of the Driving Question Board.	Display the two fertiliser formulae on the board in large font: $\text{Ca}(\text{NO}_3)_2 \cdot \text{NH}_4\text{NO}_3$ and $(\text{NH}_4)_2\text{HPO}_4$. Say verbatim: 'Look at these two formulae very carefully. Do not calculate anything yet — just observe the subscript numbers and the brackets. In your journal, write your best explanation for why each number is exactly what it is, not one more, not one less.' Allow 3 minutes of silent individual writing. Then say: 'Turn to your shoulder partner — you have 90 seconds. Partner A speaks first, Partner B listens without interrupting, then swap.' After pair discussion, cold-call 3 pairs (select one high-confidence, one mid-confidence, one uncertain learner based on observation during pair talk). For each response, ask: 'What evidence from our earlier lessons — electron arrangement or ion formation — supports that prediction?' Apply 12-second wait time after each question before taking responses. Record every distinct prediction on the DQB under 'What we think we know'. Do NOT correct predictions — affirm that all are hypotheses to be tested.	Think-Pair-Share Prediction Anchoring: By requiring learners to commit to a written prediction before instruction, this strategy activates prior knowledge from Lessons 1–4 (electron configuration and ion formation) and creates a cognitive 'gap' that motivates the observe and explain phases. Connecting predictions directly to the phenomenon (the fertiliser bag) ensures that abstract chemical concepts remain tethered to a meaningful Kenyan agricultural context throughout.	Circulate during individual writing and look for: (a) learners who connect the subscript '2' in $\text{Ca}(\text{NO}_3)_2$ to Ca forming a 2+ ion — this indicates readiness for valency derivation; (b) learners who write generic answers ('because of the elements') — these learners will need scaffolding in Phase 3. Note names of 2–3 learners in each category for targeted cold-calling in Phase 3.
Observe Phase  VIDEO: Simplifying Radicals Review Principles - Basic Source: CK-12  Search ARES for similar videos  HTML:	Activity A — Group Derivation Table (25 min): Learners work in groups of 4. Each group receives a printed 'Valency Derivation Table' (or views it on the ARES platform) listing the first 20 elements with their group numbers and electron configurations already filled in from Lesson 3. The table has three blank columns: 'Electrons lost/gained/shared', 'Ion charge',	Distribute the Valency Derivation Table. Say: 'Your task in Activity A is to complete the three blank columns using only what you already know: your electron configuration diagrams from Lesson 3 and your ion formation notes from Lesson 4. Do not look up answers — derive them.' Circulate and use these specific prompts for groups who are stuck:	Guided Inquiry with Structured Data Tables: Learners are not told valency values — they derive them from ion charges they already established in Lesson 4. This 'derivation before definition' sequence (NGSS Science and Engineering Practice 8 — Obtaining, Evaluating, and Communicating Information) ensures that valency is understood	Check each group's completed Valency Derivation Table before moving to Phase 3. Look for: (a) correct ion charges for Ca (2+), N in NH_4^+ context, and P in PO_4^{3-} context — these groups are ready for Phase 3 without scaffolding; (b) groups that confuse group number with valency for transition-like cases — provide a targeted prompt: 'Is the group number

Observation: Penny Oxidation (PLIX) Source: CK-12 Search ARES for similar readings	and 'Valency'. Groups complete the table using their Lesson 3 notes on ion formation, then circle all elements that appear in CAN or DAP. Activity B — Radical Cards (10 min): Each group receives a set of seven 'Radical Cards' (one card per KICD radical). Each card shows the radical's name, formula, and charge. Learners physically sort the cards by charge magnitude (1, 2, 3) and by whether the charge is positive or negative. They then match each radical to its appearance (if any) in CAN or DAP on a matching worksheet.	For Group 1 (if they hesitate on Group 1 elements like Na): 'Sodium is in Group 1. How many electrons does it have in its outer shell? If it loses those to become stable, what charge does it carry?' Wait 10 seconds. For Group 2 elements (Mg, Ca): 'Magnesium loses 2 electrons. What charge does Mg^{2+} carry? If the charge is $2+$, what is its valency?' Wait 12 seconds. For Group 7 (Cl): 'Chlorine gains 1 electron. What charge does Cl^- carry? What is its valency?' After 25 minutes, say: 'Stop Activity A. Do not share answers yet.' Transition to Activity B: 'I am now giving each group seven Radical Cards. These are NOT single atoms — they are groups of atoms that behave as a single unit with a fixed charge. Read each card carefully. Sort them by charge, then match them to CAN and DAP.' Allow 10 minutes. Cold-call 2 groups to explain their sorting logic. Ask: 'The nitrate radical carries a charge of $1-$. Based on your valency table, what valency does nitrate have?' Wait 12 seconds before taking answers.	as a consequence of atomic structure, not memorised as an arbitrary number. The Radical Card sorting activity provides a concrete, kinaesthetic representation of abstract charge values, and the physical act of matching radicals to CAN and DAP keeps the phenomenon central.	always the same as the valency? Look at what happens when the atom actually forms an ion.' Do not move to Phase 3 until at least 80% of groups have correctly completed 15 or more of the 20 elements.
Explain Phase  VIDEO: Simplifying Radicals Review Principles - Basic Source: CK-12 Search ARES for similar videos  HTML: Observation: Penny Oxidation (PLIX) Source: CK-12 Search ARES for similar	The teacher formalises the concept of valency and introduces oxidation numbers through three worked examples projected on the board, each rooted in a Kenyan compound. Learners then work through four practice problems in pairs, applying the criss-cross rule to derive chemical formulae. The class culminates in a whole-group derivation of $Ca(NO_3)_2$ and $(NH_4)_2HPO_4$ from first principles, directly answering the predictions made in Phase 1.	Begin with a direct instruction segment (10 minutes). Write on the board: 'VALENCY = the combining power of an element or radical, determined by the number of electrons it loses, gains, or shares.' Say: 'This is not a new idea — it is the ion charge from your table, expressed as a number without the sign.' Then write: 'OXIDATION NUMBER = the charge an atom would carry if the compound were fully ionic.' Work through Worked Example 1 — Urea ($CO(NH_2)_2$): 'Urea is used on	Worked Example — Gradual Release — Criss-Cross Application (NGSS SEP 5 — Using Mathematics and Computational Thinking): The teacher models three increasingly complex examples before releasing learners to practise independently. Each example is explicitly connected to a Kenyan compound (urea on Rift Valley maize farms, lime on Kericho tea farms, washing soda in Kenyan households), ensuring that the abstract criss-cross rule is always	During pair practice, collect the written answers for Problem 1 (ammonium sulphate) from every pair before the cold-call. Scan quickly for: (a) correct formula $(NH_4)_2SO_4$ with brackets — meets expectation; (b) formula NH_4SO_4 (missing subscript 2) — common error indicating the learner applied the valency of NH_4 as 1 but forgot to criss-cross correctly; (c) formula $N_2H_8SO_4$ (expanded the ammonium group) — indicates the learner does not yet understand that radicals stay intact. Use errors

readings		Kenyan farms, especially on maize in the Rift Valley. What is the oxidation number of carbon in urea? Oxygen is always -2, hydrogen always $+1$. Let carbon be x. Solve: $x + (-2) + 2(-3 + 2 \times 1) = 0$.' Work step by step aloud. Pause after each step and say: 'What do I do next?' Apply 10-second wait time before cold-calling. Worked Example 2 — Calcium Nitrate $\text{Ca}(\text{NO}_3)_2$: 'Calcium has valency 2. Nitrate (NO_3^-) has valency 1. Using the criss-cross rule: the valency of calcium becomes the subscript of nitrate, and the valency of nitrate becomes the subscript of calcium — but since nitrate's valency is 1, calcium's subscript is 1, and nitrate's subscript is 2. So we get $\text{Ca}(\text{NO}_3)_2$.' Write on the board step by step. Ask: 'Does this match what was on the CAN bag? What does this tell us?' Wait 12 seconds. Cold-call 3 learners. Worked Example 3 — Sodium Carbonate Na_2CO_3 (used in washing soda found in Kenyan households): 'Na has valency 1. Carbonate (CO_3^{2-}) has valency 2. Criss-cross: Na gets subscript 2, carbonate gets subscript 1. Formula: Na_2CO_3.' Now release to pair practice. Display 4 problems: (1) Ammonium sulphate (common Kenyan fertiliser) — NH_4^+ and SO_4^{2-}; (2) Calcium hydroxide (lime used on Kenyan tea farms) — Ca^{2+} and OH^-; (3) Potassium nitrate — K^+ and NO_3^-; (4) Ammonium phosphate — NH_4^+ and PO_4^{3-}. Allow 12 minutes for pair work. Circulate and whisper targeted prompts. After pair work, cold-call one pair per problem to present their answer on the board. For each answer, ask the class:	anchored to a tangible, culturally relevant substance. The final return to CAN and DAP closes the explain–phenomenon loop, giving learners the satisfaction of having derived a real-world formula from first principles.	(b) and (c) to address whole-class misconceptions before moving to Phase 4.
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		'Do you agree or disagree, and why?' Wait 12 seconds. Finally, return to DAP: say 'Now let us derive $(\text{NH}_4)_2\text{HPO}_4$ together. NH_4^+ has valency 1. HPO_4^{2-} has valency 2. Criss-cross gives $(\text{NH}_4)_2\text{HPO}_4$. This is exactly what is printed on the DAP bag in Kericho.' Pause. Say: 'Was anyone's Phase 1 prediction correct? What does it feel like to have derived a real industrial formula from atomic structure alone?' Allow 3 learners to respond.		
Driving Question Board (DQB) Creation  VIDEO: Simplifying Radicals Review Principles - Basic Source: CK-12  Search ARES for similar videos  HTML: Observation: Penny Oxidation (PLIX) Source: CK-12  Search ARES for similar readings	Learners individually revisit their Phase 1 predictions on the DQB and use a different colour marker (or a digital annotation tool on ARES) to annotate each prediction with one of three labels: CONFIRMED, REVISED, or STILL OPEN. Each learner then writes one new question that this lesson has raised for them on a sticky note and adds it to the DQB under the section 'New questions Lesson 5 has opened.' The class does a 3-minute gallery walk around the DQB before the teacher facilitates a 5-minute whole-class discussion to identify which questions are likely to be answered in Lessons 6–8 (chemical bonding, chemical equations).	Say: 'Return to the predictions you wrote at the start of this lesson on the DQB. Using a red marker (or the red annotation pen on ARES), label each prediction CONFIRMED if today's evidence supports it, REVISED if you now understand it differently, or STILL OPEN if you still cannot explain it.' Allow 4 minutes. Then say: 'Now write one genuine question — something you are still wondering about — on a sticky note and post it on the DQB.' Allow 2 minutes. Facilitate gallery walk: 'Walk around the DQB for 3 minutes. Read your classmates' new questions. Put a tick on any question you share.' After gallery walk, cold-call 3 learners whose questions received the most ticks. For each question, ask: 'Which future lesson do you think will answer this? What evidence would you need to confirm the answer?' Wait 12 seconds after each. Record the top 3 unanswered questions on the DQB in a 'Carry-forward to Lesson 6' box. Explicitly connect to the driving question: 'We now know why $\text{Ca}(\text{NO}_3)_2$ has a subscript 2 on the nitrate. But the	Annotation-Based DQB Revision (NGSS SEP 6 — Constructing Explanations): Requiring learners to publicly revise their own predictions using evidence gathered in the lesson models authentic scientific practice — scientists update their models when new evidence arrives. The 'CONFIRMED / REVISED / STILL OPEN' trichotomy makes metacognition explicit and visible, and the gallery walk builds a shared intellectual record of the class's growing understanding of the driving question.	Count the number of 'CONFIRMED' vs. 'STILL OPEN' labels on the DQB predictions. If more than 40% of predictions are still labelled 'STILL OPEN' after the explain phase, this is a signal that the criss-cross derivation in Phase 3 did not consolidate for a significant portion of the class — plan a 10-minute re-teach at the start of Lesson 6 using a different approach (e.g., colour-coded valency tiles). Also note which learners did not post a new question — these learners may be disengaged or overwhelmed and need individual check-in.

		driving question also asks why different formulae cause different plant responses. That answer lives partly in the next sub-strand — Chemical Bonding. Keep that question alive.'		
Model Building Phase  VIDEO: Simplifying Radicals Review Principles - Basic Source: CK-12  Search ARES for similar videos  HTML: Observation: Penny Oxidation (PLIX) Source: CK-12  Search ARES for similar readings	Learners retrieve their cumulative particle model diagram begun in Lesson 1 (which currently shows atomic structure, electron configuration shells, and ion formation arrows). They now add a new layer to the model: a 'Valency & Formula Key' panel in which they (1) annotate each of the first 20 elements with its valency derived today; (2) add a 'Radicals Box' listing all seven KICD radicals with name, formula, charge, and valency; (3) draw arrows from the Ca^{2+} ion and the NO_3^- radical to the derived formula $\text{Ca}(\text{NO}_3)_2$, and from NH_4^+ and PO_4^{3-} to $(\text{NH}_4)_2\text{HPO}_4$. Learners write one sentence below their model explaining how valency bridges atomic structure and the printed formula on the Kericho fertiliser bags.	Say: 'Take out your cumulative model from Lesson 1. Today we add the most powerful layer yet — the layer that connects atomic structure directly to the formula on the fertiliser bag.' Display the model template on the board showing where the new Valency and Formula Key panel should be added. Say: 'Your Radicals Box must contain all seven radicals from the KICD curriculum — name, formula, charge, and valency. Do not leave any out.' Allow 10 minutes for individual model building. Circulate and look for: (a) correct criss-cross arrows from ion to formula; (b) all 7 radicals present. After 10 minutes, say: 'Write one sentence — not a list, a full sentence — that a farmer in Kericho could understand, explaining why the formula $\text{Ca}(\text{NO}_3)_2$ cannot be CaNO_3 or Ca_2NO_3.' Cold-call 2 learners to read their sentence aloud. For each, ask: 'Is the sentence accurate? Is it something a farmer could actually use?' Wait 10 seconds. Close by saying: 'In Lesson 6, you will use these valencies and radicals to write and balance full chemical equations — and in Lessons 7 and 8, we will discover why the bonds inside these formulae determine how the compounds behave in tea farm soil. Keep this model safe.'	Cumulative Visual Model Revision (NGSS SEP 2 — Developing and Using Models): Each lesson in this sequence adds one layer to the same visual model, so that learners can literally see their understanding grow. The explicit arrow from $\text{Ca}^{2+} + \text{NO}_3^- \rightarrow \text{Ca}(\text{NO}_3)_2$ drawn by the learner themselves (rather than copied from a board) consolidates the criss-cross rule as a visual and kinaesthetic memory, not just a procedural step. The farmer-accessible sentence requirement (plain-language explanation) develops the KICD competency of Communication and Collaboration and checks for deep conceptual understanding rather than surface procedural fluency.	Collect the cumulative models at the end of the lesson (or photograph them on ARES). Use the following exit rubric: EXCEEDS — all 7 radicals correctly listed, correct criss-cross arrows for both CAN and DAP, sentence correctly links valency to formula without procedural language; MEETS — all 7 radicals listed, at least one correct criss-cross arrow, sentence mentions valency and formula; APPROACHES — 5–6 radicals listed, criss-cross arrows present but one error, sentence is vague; BELOW — fewer than 5 radicals, no criss-cross arrows, sentence absent or incorrect. Return models with written feedback before Lesson 6.

D. TEACHER REFLECTION

After the lesson, consider the following questions: (1) Were learners able to derive valency independently from their ion-formation notes, or did most groups rely on being told? If derivation was weak, revisit the ion formation notes from Lesson 4 at the start of Lesson 6 before introducing chemical equations. (2) Which of the seven KICD radicals caused the most confusion — phosphate (PO_4^{3-}) and hydrogen carbonate (HCO_3^-) are frequently conflated? If so, prepare a dedicated 5-minute radical drill at the start of Lesson 6. (3) Did the farmer-in-Kericho sentence reveal genuine conceptual understanding, or were learners reproducing phrases from the board? Genuinely explanatory sentences will use the word 'valency' or 'charge' causally (e.g., 'because calcium has a valency of 2 and nitrate has a valency of 1, you need two nitrates to balance one calcium'). (4) Were all groups able to derive $(\text{NH}_4)_2\text{HPO}_4$ from first principles, or did the HPO_4^{2-} ion (rather than PO_4^{3-}) cause confusion? Note that DAP contains HPO_4^{2-} (hydrogen phosphate), not PO_4^{3-} — if this caused confusion, acknowledge it explicitly in Lesson 6 as an example of how real industrial compounds can be more complex than ideal textbook cases, which is itself a valuable scientific lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners completed a Valency Derivation Table for the first 20 elements, deriving ion charges and valencies from their Lesson 3–4 electron configuration and ion formation notes. They sorted seven KICD Radical Cards (NH_4^+ , SO_4^{2-} , NO_3^- , OH^- , CO_3^{2-} , PO_4^{3-} , HCO_3^-) by charge and matched them to the CAN and DAP formulae from the Kericho phenomenon.
What did I learn?	Valency is the combining power of an element or radical, determined by the number of electrons lost, gained, or shared; it equals the magnitude of the ion charge. Oxidation numbers formalise the charge carried by an atom in a compound. The criss-cross rule uses valencies to generate correct chemical formulae. The seven KICD radicals are polyatomic ions that remain intact in reactions. $\text{Ca}(\text{NO}_3)_2$ has subscript 2 on nitrate because calcium's valency (2) criss-crosses with nitrate's valency (1); $(\text{NH}_4)_2\text{HPO}_4$ has subscript 2 on ammonium because phosphate's valency (2) criss-crosses with ammonium's valency (1).
How does this explain the phenomenon?	The specific subscript numbers in fertiliser formulae like CAN and DAP are not arbitrary — they are mathematical consequences of the valencies (ion charges) of the component elements and radicals, which are themselves consequences of electron arrangement in the atom. This is why a farmer in Kericho sees $\text{Ca}(\text{NO}_3)_2$ and not CaNO_3 : calcium must combine with exactly two nitrate units to achieve electrical neutrality, because calcium carries a 2+ charge and each nitrate carries a 1– charge.

LESSON 6 (80 minutes): Writing Chemical Formulae of Common Compounds

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use valencies of elements and radicals to systematically write and interpret chemical formulae of common compounds, then decode the fertiliser formulae from the anchoring phenomenon.
Knowledge	Learners write chemical formulae of common compounds using valencies of elements and radicals (ammonium ion, sulphate, nitrate, hydroxide, carbonate, phosphate, hydrogen carbonate) as required by the KICD Sub-Strand 2.1 specific learning outcome (e).
Skills	Learners apply the crossover-valency method, verify formulae by checking charge balance, decode complex fertiliser formulae (CAN and DAP), and begin constructing a class chart of common substances with names and formulae (Project 2).
Attitudes	Learners appreciate how the inner structure of the atom — specifically valency arising from electron arrangement — determines the fixed ratios in which elements combine, and why the two fertiliser bags on the Kericho tea farm carry completely different formulae despite both containing nitrogen.
Key Inquiry Question	Why is electron arrangement significant? How are elements arranged in the periodic table?
Purpose in Storyline	In Lessons 1–5 learners built up from atomic structure → electron arrangement → periodic table organisation → ion formation → valency and oxidation numbers of elements and radicals. Lesson 6 is the payoff lesson: learners now use those valencies as a tool to write chemical formulae systematically. They decode NaCl, MgO, Al ₂ O ₃ , CaCO ₃ , NH ₄ NO ₃ , and finally (NH ₄) ₂ HPO ₄ — the two fertiliser formulae that have puzzled the Kericho learner since Lesson 1. This resolves a major part of the driving question and launches Project 2.
Safety Notes	No hazardous chemicals are handled in this lesson. If coloured card or scissors are used for formula-building card activities, learners should handle scissors with care and keep the workspace tidy. No specific PPE required.

B. LESSON OVERVIEW

Learners apply the crossover-valency algorithm to write chemical formulae for a sequenced set of compounds — moving from simple binary compounds (NaCl, MgO) through compounds with radicals (CaCO₃, NH₄NO₃) to the complex fertiliser formulae (CAN and DAP) from the anchoring phenomenon. A Predict–Observe–Explain–DQB–Model structure is used throughout. By the end of the lesson learners can (a) write and verify formulae for any compound whose constituent valencies they know, (b) explain in their own words why NH₄NO₃ and (NH₄)₂HPO₄ have different formulae despite both containing nitrogen, and (c) add correctly written formulae to the class chart (Project 2). The lesson is 80 minutes and requires no laboratory chemicals — only previously prepared valency reference cards (from Lesson 5), chalk/whiteboard, learner exercise books, and a class Project 2 chart poster.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Second partial derivative test	Each learner opens their exercise book to a fresh page headed 'Lesson 6 Predictions'. They are shown five compound names	Write the five compound names clearly on the board before learners enter, or reveal them at the start of the phase. Say: 'You	Think-Pair-Share with Prediction Journaling. This activates prior knowledge from Lesson 5 (valency and oxidation numbers), creates	Teacher scans the room during silent writing. Observable indicators: (1) Are learners consulting their valency cards

Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Chemical Bonding (Flexbook) Source: CK-12 Search ARES for similar readings	written on the board: (1) Sodium chloride, (2) Magnesium oxide, (3) Calcium carbonate, (4) Ammonium nitrate, (5) Di-ammonium hydrogen phosphate. Without any instruction, each learner attempts to write what they think the chemical formula of each compound might be, using only their Lesson 5 valency reference card. After 4 minutes of individual silent writing, learners share predictions with a partner (Think-Pair-Share). Selected pairs then share aloud with the class. Learners record any disagreements between their prediction and their partner's in a coloured margin note marked '?'.	have your valency cards from Lesson 5. Without any new teaching from me, try to write the formula for each of these five compounds. You have 4 minutes — work in silence.' Set a visible timer. After 4 minutes, say: 'Turn to your neighbour. Compare your formulae. Where you disagree, circle the compound name.' Allow 2 minutes of pair discussion. Then cold-call 5 different pairs (one per compound) using the register: 'Pair Achieng and Mwangi — what did you write for sodium chloride?' After each pair responds, ask the class: 'Thumbs up if you wrote the same thing, thumbs sideways if different.' Do NOT correct any prediction yet — say: 'We are going to find out together. Keep your predictions visible — you will revisit them at the end of the lesson.' Write a tally on the board of how many different formulae were suggested for compound 5 (DAP). This creates productive cognitive dissonance.	personal stakes in the outcome, and makes misconceptions visible to the teacher before instruction begins. The visible tally of differing DAP predictions anchors the need for a systematic method.	(good — prior knowledge transfer)? (2) Are learners writing any formula at all, or leaving blanks (signals valency recall gap)? (3) Do most learners write NaCl correctly but struggle with Al_2O_3 or DAP (signals the crossover method has not yet been internalised)? Mentally note which learners leave compound 4 or 5 blank — target them for guided support in Phase 3.
Observe Phase VIDEO: Second partial derivative test Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Chemical Bonding (Flexbook) Source: CK-12 Search ARES for similar readings	Learners observe a step-by-step teacher demonstration of the crossover-valency method on the board, working through four progressively complex examples. For each example, learners follow along by writing the same steps in their exercise books. The four worked examples are: (A) NaCl — Na^{+1}, Cl^{-1}, crossover gives NaCl; (B) MgO — Mg^{2+}, O^{2-}, crossover gives MgO (introduce simplification rule — divide by HCF); (C) Al_2O_3 — Al^{3+}, O^{2-}, crossover gives Al_2O_3; (D) CaCO_3 — Ca^{2+}, CO_3^{2-} (radical bracket rule demonstrated). After each example, learners check their written formula against the board	Stand at the board and say: 'I am going to show you the four steps of the crossover method. Your job is to copy every step into your book — I will pause after each step.' Step 1: Write the symbols of the two combining species side by side with their valencies (charge numbers) written above — use + for cations and – for anions. Step 2: Cross the valency numbers diagonally, dropping the sign, to become subscripts. Step 3: Simplify the subscripts by dividing by their highest common factor. Step 4: Check — multiply each subscript by its ion's charge and confirm the total charge is zero. After example B (MgO), pause and	Worked Example Progression with Annotated Note-Taking. By moving from simplest (1:1 ratio) through simplification-needed (2:2 → 1:1) to polyatomic radicals, learners observe the pattern of increasing complexity in a controlled sequence. The structured text reading consolidates the bracket rule as a literacy-based sensemaking moment, embedding scientific vocabulary (polyatomic ion, radical, subscript) in context.	After example C (Al_2O_3), teacher circulates for 60 seconds and checks books. Observable indicator: Do learners correctly write Al_2O_3 (not Al_3O_2)? Confusion here signals the direction of crossover is not yet secure. Ask one learner who wrote it incorrectly: 'Show me where you put the 3 and where you put the 2 — talk me through it.' This targeted oral probe takes 30 seconds and informs pacing. After the structured text reading, ask: 'Hold up your book — show me the sentence you underlined.' Scan for correct identification.

	and mark it correct or incorrect. Learners then read a short 10-line structured text (displayed on the board or printed card) titled 'Why brackets matter in ionic formulae', which explains that when a polyatomic ion (radical) appears more than once, the whole group is placed in brackets with a subscript outside.	ask: 'Magnesium has valency 2 and oxygen has valency 2. The crossover gives me Mg_2O_2. What do I do next?' Wait 10 seconds. Cold-call 3 learners by name before accepting an answer. After example D (CaCO_3), say: 'The carbonate ion is a radical — it is a group of atoms that stays together. When do I need to use brackets?' Wait 12 seconds. Cold-call 2 learners. After all four examples say: 'Now read the short text on the board silently for 2 minutes, then underline the one sentence that explains when brackets must be used.'		
Explain Phase  VIDEO: Second partial derivative test Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Chemical Bonding (Flexbook) Source: CK-12  Search ARES for similar readings	Learners now apply the crossover method independently and in pairs to a graded set of six compounds. The set is structured in three tiers. Tier 1 (individual, 5 min): NaCl and MgO — binary compounds, no radicals, no simplification needed beyond what was demonstrated. Tier 2 (pairs, 7 min): CaCO_3 and Al_2O_3 — one with a radical, one with an Al^{3+} ion requiring a non-trivial crossover. Tier 3 (pairs, 8 min): NH_4NO_3 and $(\text{NH}_4)_2\text{HPO}_4$ — the fertiliser formulae from the phenomenon. For Tier 3, learners are asked to do three things: (a) write the formula using the crossover method, (b) identify every element present and its source ion, and (c) write one sentence in their exercise book answering: 'Why does DAP have a 2 in front of NH_4 but CAN does not?' The lesson then moves to a 5-minute whole-class debrief where two learners come to the board to write their Tier 3 formulae and explain their reasoning aloud.	Distribute (or reveal on board) the six compound tasks. Say: 'Tier 1 — work alone, 5 minutes. Tier 2 — work with your partner, 7 minutes. Tier 3 — this is where we finally decode the fertiliser bags from Kericho.' During Tier 1 and 2, circulate purposefully. If a learner writes Mg_2O_2 instead of MgO, say: 'You have the right crossover — now look at the subscripts. What is the HCF of 2 and 2?' and walk away to let the learner self-correct. For Tier 3, circulate to the learners identified in Phase 1 as leaving compound 5 blank and offer a scaffold: 'Let's start with the ammonium ion — what is its valency? Write it above NH_4. Now what is HPO_4's valency? Write it above HPO_4. Now cross them.' During the debrief, invite one learner to write NH_4NO_3 and another to write $(\text{NH}_4)_2\text{HPO}_4$. After each, ask the class: 'What do you notice? Are you happy with that formula? Does the charge balance?' Use the charge-balance check (Step 4 from Phase 2) publicly. Then ask: 'So — why	Tiered Practice with Phenomenon Anchoring. The three-tier structure respects the zone of proximal development — all learners achieve success at Tier 1 before being stretched. Tier 3 is explicitly framed as 'decoding the fertiliser bag', making the abstract procedure directly relevant to the real-world phenomenon. The sentence-writing task ('Why does DAP have a 2...') forces explanatory reasoning, not just procedural execution, which is the hallmark of sensemaking over rote formula writing.	During Tier 2 circulation, teacher looks for: (1) correct use of brackets for CaCO_3 — do learners write $\text{Ca}(\text{CO}_3)$ or CaCO_3? (Both are acceptable but the latter is conventional — prompt discussion of convention.) (2) Do pairs negotiate when they disagree, or does one learner simply copy the other? Observable indicator of collaboration. During Tier 3 debrief, teacher listens to the board learner's oral explanation. Specific indicator: Can the learner say 'the valency of HPO_4^{2-} is 2, so the crossover gives 2 for NH_4^{+} — not just read out the formula? If not, the teacher models the charge-balance check aloud and asks the class to verify.

		does DAP have two ammonium groups and CAN has only one?' Wait 15 seconds. Cold-call 3 learners before synthesising: 'It comes down to valency — the HPO_4^{2-} ion has a valency of 2, so it needs two NH_4^+ ions to balance the charge. That is the hidden pattern the learner in Kericho was wondering about.'		
Driving Question Board (DQB) Creation  VIDEO: Second partial derivative test Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Chemical Bonding (Flexbook) Source: CK-12  Search ARES for similar readings	Each learner revisits their Lesson 1 DQB sticky note (or their recorded driving question from their exercise book) and writes a new sticky note or journal entry with the heading 'What I can now explain'. They must write at least two complete sentences that directly address the driving question: 'Why do the fertiliser bags on Kenyan farms carry such specific and different chemical formulae — and what does the inner structure of an atom have to do with it?' Learners then physically add their new sticky notes to the class DQB under the column 'We now understand...'. A class volunteer reads three notes aloud. Learners also record one remaining question they still have — these are placed under 'We still wonder...' on the DQB.	Return learners' attention to the class DQB poster. Say: 'In Lesson 1 you wrote what you wondered. In Lessons 2 through 5 you built the tools — atomic structure, electron arrangement, ion formation, valency. Today you used those tools. So — what can you now explain about the fertiliser bags that you could not explain in Lesson 1?' Give 4 minutes of silent individual writing. Then say: 'Come and add your 'We now understand' note to the board. Keep your 'We still wonder' note too — we need it.' As learners place notes, ask a volunteer to read three aloud. After the readings, point to the phenomenon description: 'The learner in Kericho wondered why CAN and DAP look the same but have different formulae. Has anyone written an answer to that on their note?' Cold-call 2 learners whose notes address this directly. Summarise: 'The formula is not random — it is determined by valency, which comes from electron arrangement, which reflects atomic structure. The hidden pattern has a name: it is the crossover method, and it works because nature enforces charge balance.'	Driving Question Board with Progressive Claim-Building. The DQB is a public epistemic artefact that makes the class's growing understanding visible over the eight-lesson sequence. By explicitly linking today's new sticky notes to Lesson 1 questions, learners experience the satisfying arc of science: confusion → evidence → explanation. The 'We still wonder' column maintains intellectual humility and sets up Lessons 7 and 8 (balanced equations, chemical families).	Teacher reads all new sticky notes added to 'We now understand'. Observable indicator: Do learners' sentences mention valency, charge balance, or electron arrangement — or do they only describe the procedure ('you crossover the numbers')? Procedural-only responses signal that the conceptual link between atomic structure and formula-writing has not been made. These learners are flagged for a brief clarifying conversation at the start of Lesson 7.
Model Building	Learners return to their cumulative atomic model (begun in Lesson 2	Say: 'Open your cumulative model — the diagram you have been	Cumulative Model Revision with Project Integration. The cumulative	Teacher checks that the formula zone arrows connect correctly

Phase  VIDEO: Second partial derivative test Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Chemical Bonding (Flexbook) Source: CK-12  Search ARES for similar readings	and revised in each subsequent lesson). Today's revision task is to annotate the model with a new layer: a 'formula zone'. For each of two elements in their model (one metal, one non-metal), learners draw an arrow from the ion formed (annotated in Lesson 4) to a box showing the formula of the simplest compound that element forms with its most common partner — e.g., $\text{Na}^+ \rightarrow \text{NaCl}$; $\text{Mg}^{2+} \rightarrow \text{MgO}$; $\text{Al}^{3+} \rightarrow \text{Al}_2\text{O}_3$. Learners also begin their individual contribution to Project 2 by writing the names and formulae of at least four compounds (from today's set of six) neatly on their section of the class chart template. The chart template columns are: Substance Name Chemical Formula Valency of cation Valency of anion/radical Formula verified (✓/✗).	building since Lesson 2. Today we add the formula zone. Pick any one metal and one non-metal from your model. Draw an arrow from the ion you labelled in Lesson 4 into a new box. In that box, write the simplest compound formula.' Demonstrate with one example on the board: draw $\text{Na}^+ \rightarrow \text{box} \rightarrow \text{NaCl}$. Give learners 5 minutes for the model revision. Then display the Project 2 chart template on the board (or distribute printed templates). Say: 'For the last 5 minutes, fill in at least four rows of the class chart for compounds you wrote today. Every row must have the formula verified using the charge-balance check — if the charges balance, tick the verified column.' Collect or photograph the class chart at the end of the lesson to track Project 2 progress. End the lesson by saying: 'Next lesson we will use these same formulae to write and balance full chemical equations — the last step before you can read any chemical reaction the way a farmer in Kericho reads a fertiliser bag.'	model makes the lesson's conceptual contribution physically visible within the broader knowledge structure learners have been building. Adding the 'formula zone' shows learners that formulae are not isolated facts but emerge from ion charges (Lesson 4) which emerge from electron arrangement (Lesson 2). The Project 2 chart integrates procedural consolidation with a real-world artefact — a reference tool learners can actually use when reading fertiliser bags or chemical labels.	from the ion to the formula (not from the atom to the formula — the ion stage must be visible). Observable indicator: Does the learner's model show the ion (e.g., Mg^{2+}) as an intermediate step before the formula (MgO)? If a learner draws an arrow directly from the Mg atom to MgO without showing the Mg^{2+} ion, prompt: 'What happens to magnesium before it bonds with oxygen? Show me that step first.' For Project 2, check that at least one row has the verified tick and that the verification is correct (not just automatically ticked).
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your journal:

1. PHENOMENON CONNECTION — When learners wrote the sentence explaining why DAP has $(\text{NH}_4)_2$ while CAN has NH_4NO_3 , what proportion of the class connected the answer explicitly to valency and charge balance (not just to 'the crossover method')? If fewer than two-thirds did, plan a short re-engagement at the start of Lesson 7 using the fertiliser bag labels as a prompt.
2. TIER 3 STRUGGLE POINTS — Which specific compound caused the most errors: was it the HPO_4^{2-} valency (2^-), the bracket placement, or the simplification step? Adjust the opening of Lesson 7 accordingly.
3. PROJECT 2 QUALITY — Review the class chart rows completed today. Are formulae written in standard convention (e.g., CaCO_3 not $\text{Ca}(\text{CO}_3)_1$)? Are the verified ticks based on actual charge-balance checks? If the chart has errors, correct them publicly at the start of Lesson 7 before adding more rows.
4. DQB DEPTH — Read the 'We now understand' sticky notes. Count how many mention: (a) valency, (b) electron arrangement, (c) charge balance, (d) the Kericho phenomenon by name. A high-quality class response will include all four. A response that only mentions the crossover procedure signals that conceptual depth is not yet secure and the phenomenon connection needs to be revisited in Lesson 7.
5. DIFFERENTIATION — Were learners who left Tier 3 predictions blank in Phase 1 able to complete Tier 3 with scaffolding by Phase 3? If not, consider a paired

'formula buddy' system for Lesson 7, pairing those learners with peers who demonstrated strong crossover fluency today.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed a teacher-modelled demonstration of the four-step crossover-valency method applied to four progressively complex compounds (NaCl, MgO, Al ₂ O ₃ , CaCO ₃), including the bracket rule for polyatomic radicals, and read a short structured text explaining when brackets must be used in ionic formulae.
What did I learn?	Learners can apply the crossover-valency method to write and verify chemical formulae for common compounds including binary compounds and those containing radicals; they can decode the fertiliser formulae CAN [Ca(NO ₃) ₂ ·NH ₄ NO ₃] and DAP [(NH ₄) ₂ HPO ₄] from the anchoring phenomenon, and explain why the two formulae differ in terms of the valency and charge-balance requirements of their constituent ions.
How does this explain the phenomenon?	The fixed ratios in chemical formulae are not arbitrary — they are determined by the valencies of the combining ions, which arise from electron arrangement, which reflects atomic structure. The crossover-valency method encodes nature's requirement that every compound must be electrically neutral (total positive charge = total negative charge). This is why the learner in Kericho sees different formulae on her two fertiliser bags: HPO ₄ ²⁻ has a valency of 2 and therefore demands two NH ₄ ⁺ ions, while NO ₃ ⁻ has a valency of 1 and requires only one NH ₄ ⁺ — the hidden pattern is charge balance, rooted in atomic structure.

LESSON 7 (80 minutes): Balancing Chemical Equations: Writing the Language of Fertiliser Formulae

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners apply their understanding of valency and ion formation to write and balance chemical equations for reactions involving the first 20 elements, and connect the conservation of atoms in balanced equations to the fixed ratios observed in fertiliser formulae.
Knowledge	Learners will know: (i) that a balanced chemical equation has equal numbers of each type of atom on both sides; (ii) the step-by-step strategy for balancing equations using valencies; (iii) the chemical equations for formation of sodium chloride (NaCl), magnesium oxide (MgO), calcium nitrate $\text{Ca}(\text{NO}_3)_2$, and ammonium phosphate $(\text{NH}_4)_2\text{HPO}_4$ from their constituent elements/compounds; (iv) that the law of conservation of mass underpins balanced equations and explains why fertiliser formulae carry fixed ratios.
Skills	Learners will be able to: (i) write word equations and then symbol equations for chemical reactions; (ii) apply valency rules to determine correct formulae of products; (iii) use the step-by-step balancing strategy (identify → count → balance by inspection → verify) to balance equations; (iv) check peers' equations systematically; (v) explain how a balanced equation encodes the conserved atom ratios seen in fertiliser formulae such as CAN and DAP.
Attitudes	Learners will: (i) appreciate that the fixed chemical formulae on fertiliser bags are a direct consequence of conserved atomic ratios governed by valency; (ii) show respect for peers' reasoning during pair checking; (iii) demonstrate responsibility and precision in counting atoms; (iv) value the connection between classroom chemistry and the agricultural practices of Kenyan farmers.
Key Inquiry Question	Why is electron arrangement significant? How are elements arranged in the periodic table?
Purpose in Storyline	In previous lessons learners explored atomic structure, periodic trends, ion formation, electron configuration, valency, oxidation numbers, and chemical formulae. Lesson 7 is the culminating evidence-gathering step: learners now write full balanced equations and discover that every balanced equation is a statement of conservation — precisely the hidden pattern the Kericho learner could not yet explain when she noticed that CAN and DAP carry different but fixed formulae. By balancing equations for the very compounds found in those fertiliser bags, learners gather the final evidence needed to answer the driving question before the synthesising Lesson 8.
Safety Notes	No hazardous chemicals are handled in this lesson. All activities are paper-based and dry-erase/chalk-board work. Ensure adequate desk space for pair work. If printed formula cards are used, ensure scissors are handled safely. Learners with visual impairments should be paired with a sighted peer and provided with large-print atom-count tables.

B. LESSON OVERVIEW

Lesson 7 is an 80-minute, paper-based sensemaking lesson in which Grade 10 learners consolidate their valency knowledge (Lessons 4–6) by learning to write and balance chemical equations. The anchoring phenomenon — the Kericho learner's puzzlement over the fixed formulae on CAN and DAP fertiliser bags — is revisited explicitly: learners discover that balanced equations are the formal proof that atoms are conserved in fixed ratios, which is exactly why fertiliser formulae cannot be arbitrary. The lesson moves through five structured phases: (1) a paired prediction task in which learners attempt to balance a familiar equation before instruction; (2) a guided observation of a step-by-step balancing demonstration using Uganda-to-Nairobi maize transport as a ratio analogy; (3) a paired explain-and-practice session covering four target equations (NaCl, MgO, $\text{Ca}(\text{NO}_3)_2$, and $(\text{NH}_4)_2\text{HPO}_4$); (4) a Driving Question Board update; and (5) a cumulative model revision in which learners annotate their atomic-structure diagrams to show conserved ratios. Peer checking with a structured checklist is central to the lesson design. KICD core competencies developed include Critical Thinking and Problem Solving, Communication and Collaboration, and Self-efficacy. Values reinforced are Respect, Responsibility, and Unity.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Video: Balancing Chemical Equations Source: PhET Interactive Simulations (English) Search ARES for similar videos  HTML: Chemical Equations: Balancing Chemical Reactions (PLIX) Source: CK-12 Search ARES for similar readings	Each learner receives a half-page 'Predict Card' showing an unbalanced equation: $\text{Na} + \text{Cl}_2 \rightarrow \text{NaCl}$ (sodium reacting with chlorine gas to form table salt). Without any instruction, learners individually attempt to balance the equation by writing coefficients. They then record: (a) their balanced equation attempt, (b) one reason WHY they think atoms must be equal on both sides, and (c) a prediction about whether the fertiliser formulae CAN $[\text{Ca}(\text{NO}_3)_2 \cdot \text{NH}_4\text{NO}_3]$ and DAP $[(\text{NH}_4)_2\text{HPO}_4]$ could ever be written differently — e.g. could DAP be $(\text{NH}_4)_3\text{PO}_4$ instead? Learners hold up their Predict Cards so the teacher can do a quick visual survey before discussion.	T distributes Predict Cards face-down. Say exactly: 'Do NOT turn the card over until I say go. I want your honest first attempt — there are no marks for this, only thinking.' [Pause 5 seconds.] 'Turn over — begin. You have 5 minutes working alone.' Circulate silently for 3 minutes, noting on a clipboard: (i) how many learners are writing $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$ (correct), (ii) how many write $\text{Na} + \text{Cl}_2 \rightarrow \text{NaCl}$ (unbalanced), (iii) how many leave blank. After 5 minutes: 'Pair with your neighbour and compare your attempts. You have 2 minutes.' Then cold-call 3 pairs: 'Baraka and Achieng, what did your pair write and why?' [WAIT 12 seconds after each cold-call before prompting.] Do NOT confirm or correct yet. Say: 'Hold your predictions — we will test them in the next phase.' Post a class tally on the board: '___ pairs think balanced, ___ pairs unsure.' This tally creates productive cognitive tension that will be resolved in the Explain phase.	PREDICT-THEN-COMPARE — Students externalise prior knowledge before instruction, making implicit valency understanding (from Lessons 4–6) visible. Comparing with a partner surfaces disagreements that motivate careful observation in Phase 2. The fertiliser formula prediction task directly links the new skill (balancing) back to the driving phenomenon.	Observable indicator: Teacher scans Predict Cards and tallies — if fewer than 30% of learners attempt a coefficient strategy (rather than changing subscripts), teacher flags this as a prerequisite gap and addresses it explicitly at the start of Phase 3 ('Coefficients change HOW MANY molecules, subscripts change the formula itself — we NEVER change subscripts when balancing.').
Observe Phase  VIDEO: Video: Balancing Chemical Equations Source: PhET Interactive Simulations (English) Search ARES for similar videos  HTML: Chemical	The teacher leads a whole-class demonstration of the 4-step balancing strategy using a maize-lorry analogy drawn on the board, then applies the strategy live to the $\text{Na} + \text{Cl}_2$ equation from the Predict Card. Learners observe and take structured notes on a pre-printed 'Observation Frame' with four columns: Step What I do Sodium atoms Chlorine atoms. After the demonstration, learners watch the teacher work through a second equation ($\text{Mg} + \text{O}_2 \rightarrow$	Draw on the board: 'ANALOGY — A lorry from Eldoret carries maize sacks to Nairobi. If 1 sack holds 2 bags of maize (like Cl_2 holds 2 Cl atoms), how many lorries do I need to deliver exactly 2 bags?' Use arrows to show: 1 lorry $\times$ 2 bags/lorry = 2 bags. Say: 'In chemistry, we call the number of lorries the COEFFICIENT. We can change how many lorries — we NEVER cut the lorry in half [cross out the subscript dramatically].' Now move to the equation. Write:	ANALOGY + THINK-ALLOUD MODELLING — The Eldoret-Nairobi maize-lorry analogy makes the abstract concept of coefficients vs. subscripts concrete and Kenyan-relevant. The teacher's live think-aloud externalises expert reasoning, giving learners a cognitive scaffold. The structured Observation Frame focuses attention on atom counts rather than superficial equation-writing, building the counting precision needed in Phase 3.	Observable indicator: During the gallery walk, teacher reads 3–4 'questions still have' entries from Observation Frames. If more than 40% of learners write questions about WHAT a coefficient means (not HOW to use it), teacher opens Phase 3 with a 2-minute re-teach of the lorry analogy using a different example (githeri recipe: 2 cups maize + 1 cup beans $\rightarrow$ 1 pot githeri; doubling the recipe requires doubling ALL ingredients, not re-defining a cup).

Equations: Balancing Chemical Reactions (PLIX) Source: CK-12 Search ARES for similar readings	MgO) using the same 4 steps, this time narrating each step aloud. Learners fill in their Observation Frame independently as the teacher works. A short 3-minute 'gallery walk' of four pre-written large-format equation cards pinned around the room (one for NaCl, MgO, Ca(NO₃)₂, and (NH₄)₂HPO₄ — each showing ONLY the unbalanced left side) allows learners to read the equations and note atom counts before the Explain phase.	$\text{Na} + \text{Cl}_2 \rightarrow \text{NaCl}$. Say aloud: 'Step 1 — Write the unbalanced equation. Done.' 'Step 2 — Count atoms on each side: left: 1 Na, 2 Cl. Right: 1 Na, 1 Cl. Not equal — Cl is short on the right.' 'Step 3 — Balance by inspection: put coefficient 2 in front of NaCl $\rightarrow \text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$. Now recount: right has 2 Na, 2 Cl — but left only has 1 Na.' 'Step 4 — Adjust: put 2 in front of Na $\rightarrow 2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$. Verify: 2 Na = 2 Na ✓ 2 Cl = 2 Cl ✓ BALANCED.' Cold-call 2 learners to echo the 4 steps back: 'Njeri, tell me Step 2 in your own words.' [WAIT 12 seconds.] 'Otieno, what do we NEVER change and why?' [WAIT 12 seconds.] Then repeat the 4-step process for $\text{Mg} + \text{O}_2 \rightarrow \text{MgO}$, narrating slowly. After gallery walk: 'In your Observation Frame, write ONE question you still have about balancing. We will answer those questions in the next phase.'		
Explain Phase  VIDEO: Video: Balancing Chemical Equations Source: PhET Interactive Simulations (English) Search ARES for similar videos  HTML: Chemical Equations: Balancing Chemical Reactions (PLIX) Source: CK-12 Search ARES for similar readings	Learners work in stable pairs. Each pair receives a 'Balancing Worksheet' containing four target equations linked to the fertiliser phenomenon: (1) $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$ [table salt — review]; (2) $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$ [magnesium oxide — common in lime fertiliser]; (3) $\text{Ca} + 2\text{HNO}_3 \rightarrow \text{Ca}(\text{NO}_3)_2 + \text{H}_2$ [calcium reacting with nitric acid to form calcium nitrate, a component of CAN]; (4) $3(\text{NH}_4)\text{OH} + \text{H}_3\text{PO}_4 \rightarrow (\text{NH}_4)_3\text{PO}_4 + 3\text{H}_2\text{O}$ [ammonium phosphate — related to DAP]. For each equation, the worksheet provides the unbalanced form, a blank atom-count table, and a space for the balanced equation. Pairs work through equations using the 4-step strategy. After completing each equation, Partner	Distribute worksheets and checklists. Say: 'You have 18 minutes for all four equations. Partner A starts as the explainer for equation 1. Partner B uses the checklist. Swap for equation 2. Ready — go.' Circulate purposefully using a 3-column tracking table (pair name equation stuck on error type). Every 4–5 minutes, pause the class for a 'Cold-Call Check': choose one pair who made an error (identified during circulation) and ask ONE partner to narrate Step 2 for that equation. Say: 'Kamau, read your atom count for the LEFT side of equation 3 — just the numbers.' [WAIT 12 seconds.] Then ask the class: 'Hands up if your count for Ca on the left is the	STRUCTURED PAIR WORK WITH RECIPROCAL TEACHING — Partners alternate between explainer and checker roles, which forces both to engage with reasoning rather than just copying answers. The Peer-Check Checklist externalises the quality criteria, making self-regulation explicit. The DAP annotation task directly applies the balancing skill to the driving phenomenon, closing the explanatory loop opened in Lesson 1.	Observable indicators: (i) Teacher's tracking table — if more than 3 pairs are stuck on the same step for equation 3 (calcium nitrate), teacher pauses the class and reteaches polyatomic ion counting: 'When you have NO₃⁻ as a unit, count the whole bracket first, THEN multiply by the subscript outside.' (ii) Peer-Check Checklists are collected at the end — if more than 25% of checklists show 'Subscripts unchanged? NO,' teacher opens the DQB phase with a targeted question: 'What is the difference between 2NaCl and NaCl₂?' as a whole-class sorting task.

	A explains the reasoning aloud while Partner B checks using a 'Peer-Check Checklist' (3 items: ✓ Subscripts unchanged? ✓ All atom types counted? ✓ Both sides equal?). Partners swap roles for each new equation. After all four are complete, each pair annotates the DAP formula on their Predict Card: 'The ratio $(\text{NH}_4)_3\text{PO}_4$ comes from N valency __, P valency __, therefore __ ammonium ions per phosphate ion.' The teacher links this directly back to the DAP bag label.	same as Kamau's.' Use discrepancies to re-teach on the spot without embarrassing the pair: 'Interesting — we have two different counts. Let's check together.' After 18 minutes: 'Annotate your Predict Card now — 3 minutes.' Then cold-call 2 pairs to share their DAP ratio annotation. Ask: 'So could DAP ever be written $(\text{NH}_4)_4\text{PO}_4$?' [WAIT 15 seconds.] Target response: 'No — because phosphate has valency 3 and ammonium has valency 1, so we always need exactly 3 ammonium ions to balance one phosphate ion — the ratio is fixed by valency.' Affirm: 'That fixed ratio IS why the bag in Kericho has that exact formula — and why it cannot be anything else.'		
Driving Question Board (DQB) Creation  VIDEO: Video: Balancing Chemical Equations Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: Chemical Equations: Balancing Chemical Reactions (PLIX) Source: CK-12  Search ARES for similar readings	The class gathers at the Driving Question Board. Each pair writes one new 'We now understand...' sticky note (green) and one new 'We still wonder...' sticky note (yellow) and places them on the board under the correct column. The teacher reads selected notes aloud and asks the class to vote — thumbs up if they share that understanding, thumbs sideways if they are still unsure. The board is reorganised so that resolved questions (answered across Lessons 1–7) are grouped on the left, with remaining open questions clustered on the right to set the agenda for Lesson 8. The driving question itself ('Why do fertiliser bags carry such specific formulae...?') is read aloud as a full class. Learners do a 'confidence thermometer' — fist (0%) to five fingers (100%) — to show how fully they can now	Hand out sticky notes. Say: 'In exactly 3 minutes, each pair writes one green and one yellow note — no more. Green = something you NOW understand that you did not before today. Yellow = something you STILL cannot explain.' After posting, read 3 green notes aloud, then ask: 'Who can add evidence from today's balancing work to support this understanding?' Cold-call 2 learners. [WAIT 12 seconds each.] Then read 2 yellow notes and say: 'These are our agenda for Lesson 8. Keep them visible.' Finally: 'Driving question — everybody read it silently... now show me your confidence thermometer.' Survey the class and note distribution. Say: 'Most of you are at 3–4 fingers. After Lesson 8, I expect everyone at 5. What is the ONE thing still blocking you?' Cold-call 1 learner who is at 2 fingers: '[Name], what	DRIVING QUESTION BOARD CURATION — The DQB makes the cumulative growth of understanding visible and metacognitive. The confidence thermometer gives the teacher real-time data on the class's explanatory readiness and signals to learners that understanding is a trajectory, not a binary. Clustering resolved vs. open questions creates a visual narrative of the evidence-gathering sequence.	Observable indicator: If fewer than 50% of learners show 3 or more fingers on the confidence thermometer, teacher identifies the dominant yellow-note theme and adds a targeted review task to the opening of Lesson 8. If the dominant yellow theme is about WHY valency determines ratios (not just HOW to balance), teacher prepares a Lesson 8 opening model-building task that connects electron configuration → valency → ionic ratio → formula → balanced equation as a single chain.

	answer it.	is the gap?' [WAIT 15 seconds.] Use the response to frame Lesson 8 preview.		
Model Building Phase  VIDEO: Video: Balancing Chemical Equations Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: Chemical Equations: Balancing Chemical Reactions (PLIX) Source: CK-12  Search ARES for similar readings	Each learner returns to their cumulative atomic-structure diagram (built and revised across Lessons 1–6). Today they add a new annotated layer: for each of the four equations practised in Phase 3, they draw a simple particle-level representation showing atoms on the left side and product on the right side, with coefficients shown as repeated particle icons (e.g., two Na atoms + one Cl₂ molecule → two NaCl units). Below their diagram they complete the sentence stem: 'The formula on the CAN bag is fixed because _____ and this is proven by the balanced equation _____, which shows that for every one calcium ion there must be exactly two nitrate ions because calcium has valency ____ and nitrate has valency ____.' The completed model is initialised by their partner as a peer-witnessed record.	Say: 'Open your cumulative model — the one you have been building since Lesson 1. Today you are adding the final evidence layer before our synthesis lesson. You have 8 minutes. Work individually but your partner will initial it at the end to confirm it is complete.' Circulate and target learners whose models lack the particle-level representation (most common gap). Ask: 'Show me two sodium atoms as circles — now show me one Cl₂ — how many circles on the right side of your arrow?' [WAIT 10 seconds for learner to draw.] In the final 2 minutes: 'Hold up your model. Partner — check the sentence stem. If it is complete and correct, initial it.' Cold-call one learner to read their completed sentence stem aloud to the class. Affirm explicitly: 'That sentence is the answer to the Kericho learner's puzzle. She now knows why the bag says Ca(NO₃)₂ and not Ca(NO₃)₃ — the balanced equation proves it.' Close: 'In Lesson 8 we put everything together — atomic structure, periodic trends, valency, formulae, and balanced equations — to fully explain both fertiliser bags. Bring your complete cumulative model.'	CUMULATIVE MODEL REVISION — Adding a particle-level visual layer to an existing model reinforces that science knowledge is iterative and cumulative. The sentence stem requires learners to chain the causal argument (electron configuration → valency → ionic ratio → formula → balanced equation → fertiliser label), making the explanatory chain explicit and testable. Peer initialling creates social accountability for completeness.	Observable indicator: Teacher scans sentence stems during circulation. Success criterion: the stem correctly names valency values (Ca = 2, NO₃⁻ = 1) and correctly states the 1:2 ratio. If more than 4 learners write incorrect valency values, teacher corrects immediately and writes the valency table on the board as a reference, reminding learners to keep it in their notes for Lesson 8.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) What proportion of pairs successfully balanced all four equations without teacher intervention? If fewer than 60%, identify which equation type caused most errors (likely equation 3 — polyatomic ions in Ca(NO₃)₂) and plan a 5-minute targeted reteach at the start of Lesson 8 using a 'bracket-first' counting rule. (2) Did learners make the explicit connection between a balanced equation and the fixed fertiliser formula, or did they treat balancing as a mechanical arithmetic exercise? Evidence: read the sentence stems in the cumulative models. If the causal chain is missing (i.e., learners wrote the correct numbers but

not the 'because valency...' reasoning), plan a Lesson 8 opening question: 'Could a farmer use a fertiliser called $\text{Ca}(\text{NO}_3)_3$? Why or why not?' (3) Was the maize-lorry analogy effective for all learners, or did some learners in the class find it confusing? Consider substituting a githeri recipe analogy (2 cups maize + 1 cup beans = 1 pot githeri; 4 cups maize + 2 cups beans = 2 pots) for any learners who were not reached by the transport analogy. (4) Review the DQB yellow sticky notes: are the remaining open questions conceptual (why valency = electron shell behaviour) or procedural (how to balance complex equations)? Conceptual gaps require the Lesson 8 synthesis to explicitly revisit electron configuration; procedural gaps can be addressed with a 10-minute practice extension. (5) Note which pairs demonstrated strong peer-checking discipline and which pairs signed off checklists without genuinely verifying — consider restructuring pairs for Lesson 8 to mix strong and developing checkers.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed the teacher model a 4-step balancing strategy (write → count → balance by inspection → verify) applied live to $\text{Na} + \text{Cl}_2 \rightarrow \text{NaCl}$ and $\text{Mg} + \text{O}_2 \rightarrow \text{MgO}$, using a maize-lorry analogy to distinguish coefficients from subscripts. They also observed four unbalanced equation cards in a gallery walk, noting atom counts for NaCl , MgO , $\text{Ca}(\text{NO}_3)_2$, and $(\text{NH}_4)_2\text{HPO}_4$.
What did I learn?	Learners learned that: (i) a balanced chemical equation has the same number of each type of atom on both sides; (ii) only coefficients (not subscripts) may be changed when balancing; (iii) the 4-step strategy (write → count → balance → verify) can be applied to equations involving elements and compounds of the first 20 elements; (iv) balanced equations prove that atoms combine in fixed ratios determined by valency — which is exactly why the fertiliser formulae CAN $[\text{Ca}(\text{NO}_3)_2 \cdot \text{NH}_4\text{NO}_3]$ and DAP $[(\text{NH}_4)_2\text{HPO}_4]$ carry those specific and unalterable chemical formulae.
How does this explain the phenomenon?	Learners explained — through paired practice, peer checking, DQB annotations, and cumulative model sentence stems — that the fixed ratios in fertiliser formulae are not arbitrary: they are a direct consequence of the valencies of the constituent elements and polyatomic ions, and these valencies arise from the electron arrangements of atoms (established in Lessons 2–4). A balanced equation is the formal, conservation-law-backed proof of those ratios. The Kericho learner's puzzle is therefore answered: the inner structure of the atom (electron configuration → valency) determines what formula a compound gets, and balancing the equation confirms that the ratio is the only one consistent with the law of conservation of mass.

LESSON 8 (40 minutes): Model Building, Project Showcase and Driving Question Board Resolution: From Electron Arrangement to Fertiliser Formula

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	By the end of this lesson, learners will consolidate the entire unit by connecting electron arrangement to valency, valency to chemical formula, and chemical formula to the labels on Kenyan fertiliser bags, demonstrating mastery through project presentations and a full resolution of the driving question.
Knowledge	Learners will demonstrate that: (i) electron arrangement in the first 20 elements determines the number of outer-shell electrons; (ii) outer-shell electrons determine valency and ion charge; (iii) valency rules and radical charges determine chemical formulae; (iv) balanced equations confirm the fixed atom ratios encoded in those formulae; (v) the specific formulae on CAN and DAP fertiliser bags are a direct consequence of the inner structure of the atoms involved.
Skills	Learners will: present electron-configuration models constructed from locally available materials; explain compound-formula charts to peers using correct chemical vocabulary; synthesise evidence from Lessons 1 through 7 into a coherent explanation; evaluate peers using a structured rubric; and resolve the driving question in writing with supporting evidence.
Attitudes	Learners will appreciate that chemistry is not abstract — it underpins Kenyan agriculture, pharmacy, and manufacturing; show confidence in presenting their own scientific reasoning; and demonstrate respect for peers during project showcase.
Key Inquiry Question	Why is electron arrangement significant? How are elements arranged in the periodic table? Together these resolve to: How does the inner structure of an atom determine what chemical formula a compound carries — and why do fertiliser bags on Kenyan farms list those exact formulae?
Purpose in Storyline	This is the culminating lesson in which all threads — atomic structure, electron configuration, periodic law, ion formation, valency, radicals, formula writing, and balanced equations — are woven into a single answer to the driving question. The Kericho learner who wondered why CAN and DAP look the same yet differ chemically now receives a full, evidence-based explanation built by the class over eight lessons.
Safety Notes	No hazardous chemicals are used. Model materials such as bottle tops, wire, and clay are safe but remind learners to handle wire ends carefully to avoid scratches. Ensure the classroom floor is clear during the showcase to prevent tripping on models.

B. LESSON OVERVIEW

Lesson 8 is a 40-minute consolidation and celebration lesson structured around two student projects and the formal resolution of the Driving Question Board. In the Predict Phase (5 minutes) learners revisit their original DQB predictions from Lesson 1 and rate how much they can now explain. In the Observe Phase (10 minutes) groups present Project 1 (3-D electron-configuration models of selected elements) and Project 2 (compound-formula charts). In the Explain Phase (12 minutes) learners collaboratively construct the full causal chain: electron arrangement leads to valency, valency determines formula, formula is encoded on the fertiliser bag. In the DQB Resolution Phase (8 minutes) the class formally answers the driving question on the board and records remaining wonderings. In the Model Building Phase (5 minutes) learners complete individual written reflections linking the unit to careers in Kenyan agriculture, pharmacy, and chemical manufacturing.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Simplifying Radicals Review Principles - Basic Source: CK-12  Search ARES for similar videos  HTML: Electron Cloud Atomic Model (Flexbook) Source: CK-12  Search ARES for similar readings	Learners open their exercise books to the predictions they recorded in Lesson 1 about why CAN and DAP have different formulae. Working individually for 2 minutes, they re-read their original prediction and privately score it: 1 = I had no idea then, 3 = I was partially right, 5 = I can now explain it fully. They then share their score with a partner and briefly say what piece of evidence from Lessons 1 to 7 changed their thinking the most. The teacher displays the original fertiliser images and the driving question on the board to anchor the reflection.	The teacher says: 'Before we share all the work you have done, I want you to travel back to Lesson 1. Read your first prediction about the driving question.' WAIT TIME: 60 seconds of silent reading. Teacher then says: 'Give yourself an honest score from 1 to 5 — not for effort, but for how far your explanation has grown. Be specific: what one lesson or activity moved the needle for you most?' WAIT TIME: 90 seconds of pair talk. Teacher circulates, listening for references to electron arrangement, valency, or ion formation to gauge readiness. Teacher probes one or two pairs: 'What exactly about electron arrangement was the turning point?' Teacher does not resolve the question yet — tension is held deliberately.	Self-assessment against prior prediction (metacognitive monitoring). The gap between the Lesson 1 prediction and the current understanding becomes the motivating frame for the showcase.	Teacher scans self-scores and listens to pair talk to identify learners who still conflate subscripts with valency or who cannot name the causal chain. These learners are mentally noted for targeted support during the Explain Phase.
Observe Phase  VIDEO: Simplifying Radicals Review Principles - Basic Source: CK-12  Search ARES for similar videos  HTML: Electron Cloud Atomic Model (Flexbook) Source: CK-12  Search ARES for similar readings	Groups of four take turns presenting their two projects at their desks or at the front (teacher decides based on class size). Project 1: each group holds up their 3-D model of an element or set of elements (e.g., Na, Mg, Al, Cl, Ca built from bottle tops for electrons, wire for shells, and clay for the nucleus). They state: the element name, atomic number, electron arrangement, and the ion it forms. Project 2: each group reads two rows from their compound-formula chart, stating compound name, formula, and the valency logic behind it — including at least one fertiliser compound (NH ₄ NO ₃ or (NH ₄) ₂ HPO ₄). Audience members use a peer-feedback card with three prompts: one thing explained clearly, one	Teacher says: 'You are not just presenting — you are teaching the class. Your audience should be able to repeat your reasoning after you finish.' During presentations teacher uses structured observation, listening for: correct electron notation (e.g., 2,8,1 for Na), correct ion charge linked to group number, and correct use of the cross-multiplication rule for formulae. After each group the teacher asks the audience: 'What valency rule did that group use to arrive at that formula? Thumbs up if you can state it.' WAIT TIME: 20 seconds. If a group mis-states a formula, teacher says: 'Interesting — class, let us check that together. What does the valency of calcium tell us about how many nitrate groups it attracts?' rather	Gallery-style peer teaching with structured feedback cards. Observing multiple models of the same concept (electron shells represented differently by each group) deepens pattern recognition across representations.	Teacher collects one peer-feedback card per group after all presentations. Errors flagged on cards inform the Explain Phase. Teacher also notes whether groups can articulate the formula of CAN or DAP using valency language — this is the core unit goal.

	chemical term used correctly, one question I still have. Presentations are kept to 90 seconds per group to fit the time budget.	than correcting directly.		
Explain Phase  VIDEO: Simplifying Radicals Review Principles - Basic Source: CK-12  Search ARES for similar videos  HTML: Electron Cloud Atomic Model (Flexbook) Source: CK-12  Search ARES for similar readings	The whole class co-constructs a causal chain on the board using five labelled boxes connected by arrows. Box 1: Atomic Structure (protons, neutrons, electrons). Box 2: Electron Arrangement (e.g., Ca is 2,8,8,2). Box 3: Valency and Ion Charge (Ca loses 2 electrons, becomes Ca^{2+}, valency 2). Box 4: Chemical Formula (cross-multiplication of valencies gives $\text{Ca}(\text{NO}_3)_2$ in CAN and $(\text{NH}_4)_2\text{HPO}_4$ in DAP). Box 5: The Fertiliser Bag Label (the formula printed on the bag is chemically necessary — it cannot be any other ratio). Each box is completed by a different learner volunteer. The class then does a two-minute silent written response: 'Using the five-box chain, explain in three sentences why DAP contains two ammonium groups for every one phosphate group.' They share one sentence with the class. The teacher then asks: 'So why do plants respond differently to CAN and DAP even though both contain nitrogen?' Learners connect the presence of phosphate in DAP (supporting root development) to the formula — which exists because PO_4^{3-} has a valency of 3 while NH_4^+ has a valency of 1, requiring two ammonium ions to balance one phosphate.	Teacher says: 'We are going to build the answer to our driving question together — one box at a time. Each box must be filled with evidence from a specific lesson.' Teacher assigns boxes to volunteers based on formative data: learners who scored themselves 1 or 2 in the Predict Phase are given Box 1 or 2 (more concrete) and supported with a prompt card. Teacher says after Box 3 is filled: 'Pause. Everyone check: does the ion charge on the board match the group number of that element in the periodic table? 30 seconds.' WAIT TIME: 30 seconds. After Box 5 is filled, teacher says: 'The formula on that fertiliser bag in Kericho is not random. It is a consequence of something invisible — the arrangement of electrons inside each atom. That is the answer to our driving question. But I want to hear it in your words.' WAIT TIME: 90 seconds of individual writing before sharing.	Collaborative causal-chain construction on the board (making the logic visible and sequential). The five-box model serves as a public summary artefact that learners can photograph or copy. Written consolidation (three-sentence response) cements individual understanding.	Teacher reads three-sentence responses over learners shoulders during writing time. Checks for: correct use of the term valency, correct reference to ion charge, and correct formula for either CAN or DAP. Learners who write only surface descriptions (both have nitrogen) receive a prompt: 'What determines how many nitrogen atoms appear in the formula?'
Driving Question Board (DQB) Creation  VIDEO: Simplifying	The class returns to the Driving Question Board posted since Lesson 1. Learners review all the sticky notes added over eight lessons — predictions, questions, and partial answers. Working as a	Teacher says: 'When we started, some of you wrote that the formulae were just random names made up by companies. Look at that note now. Has chemistry changed your mind?' WAIT TIME:	Collaborative DQB review and personal written resolution. Sorting notes into categories makes epistemic progress visible. The transition from wondering to explaining models authentic	Two-sentence sticky notes serve as a brief summative performance indicator. Teacher checks: does the learner name electron arrangement as the root cause? Does the learner link valency to

Radicals Review Principles - Basic Source: CK-12 Search ARES for similar videos  HTML: Electron Cloud Atomic Model (Flexbook) Source: CK-12 Search ARES for similar readings	whole class with the teacher facilitating, they sort remaining notes into three categories: Answered, Partially Answered, and New Wonderings. A class representative reads the driving question aloud: 'Why do the fertiliser bags on Kenyan farms carry such specific and different chemical formulae — and what does the inner structure of an atom have to do with it?' Each learner writes their personal final answer on a new sticky note in exactly two sentences and places it on the board under Answered. Three or four New Wonderings are selected and written on a fresh section of the board — for example: 'How do farmers in Kisumu know which fertiliser to use at which growth stage?', 'What happens to the ions once the fertiliser dissolves in the soil?' and 'Are there other compounds not in the first 20 elements that are used in Kenyan fertilisers?' The teacher affirms that new questions are a sign of genuine scientific thinking.	30 seconds of silent reflection. Teacher then says: 'Write your two-sentence final answer. It must include the words electron arrangement and valency. You have 90 seconds.' WAIT TIME: 90 seconds. After sticky notes are placed, teacher reads two or three aloud — choosing one strong example and one that is almost there — and asks the class: 'What could we add to this one to make it complete?' Teacher closes the DQB activity by saying: 'Science does not end with answers — it ends with better questions. The wonderings you have written here are real questions that Kenyan agricultural chemists work on every day.'	scientific discourse.	the specific formula on the bag? Notes with gaps are flagged for follow-up in subsequent sub-strands or remediation.
Model Building Phase  VIDEO: Simplifying Radicals Review Principles - Basic Source: CK-12 Search ARES for similar videos  HTML: Electron Cloud Atomic Model (Flexbook) Source: CK-12 Search ARES for similar readings	In the final 5 minutes learners complete an individual written reflection in their exercise books using the prompt: 'Imagine you are advising a tea farmer in Kericho on why the formula of CAN cannot be changed arbitrarily. Write three to four sentences using what you have learned about electron arrangement, ion formation, valency, and balanced equations to justify why $\text{Ca}(\text{NO}_3)_2 \cdot \text{NH}_4\text{NO}_3$ is the only chemically correct formula for that compound.' They then answer a careers-connection question: 'Name one career in Kenya that depends on understanding chemical formulae,	Teacher says: 'This final reflection is your voice — not a group voice. Use the five-box chain we built together as your guide.' Teacher circulates and offers a sentence starter to learners who are stuck: 'Because calcium is in Group 2, it has an outer shell with ... electrons, which means it forms an ion with a charge of ...' Teacher affirms correct use of the causal chain. In the last minute, teacher says: 'The learner in Kericho who noticed the two fertiliser bags started this journey. You have now built the scientific explanation she was looking for. That is the power of chemistry.' Teacher assigns any	Individual written application to a real Kenyan agricultural context (transfer task). The careers-connection prompt links disciplinary knowledge to socioeconomic relevance, reinforcing the ARES attitude strand and CBE life-skills integration.	Teacher collects or spot-checks reflections at the door as learners leave (exit-ticket style). The three-to-four sentence justification reveals whether learners have achieved the unit SLO at a transfer level — applying the causal chain to a novel audience (the farmer) rather than just recalling it.

	and write one sentence explaining how this unit knowledge applies to that career.' Options discussed briefly include: agricultural chemist, pharmacist, chemical engineer at the Mombasa refinery, food scientist, or secondary school chemistry teacher. Learners who finish early write a second career example.	remaining project write-up as homework if not fully completed in class.		
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D. TEACHER REFLECTION

After the lesson, the teacher considers: Did all groups demonstrate in their project presentations that they can link electron arrangement to formula — or did some groups focus only on memorised formulae without explanation? Were the two-sentence DQB final answers causal (electron arrangement causes valency causes formula) or merely descriptive (CAN has calcium and nitrogen)? Which learners still conflate the subscript in a formula with the number of atoms of the element rather than with the valency ratio? How might the New Wonderings on the DQB be used as entry points for Sub-strand 2.2 or for cross-curricular work with Agriculture? Were locally available materials (bottle tops, wire, clay) sufficient for meaningful model building, or does the school need to source additional resources before the next cohort runs this unit? The teacher also notes whether the careers discussion was substantive or rushed — if rushed, a dedicated five-minute careers integration moment should be built into Lesson 7 next time.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed electron-configuration models built from bottle tops, wire, and clay representing the first 20 elements; compound-formula charts showing names, formulae, and valency justifications for Kenyan compounds including CAN and DAP; and the completed Driving Question Board with sticky-note predictions from Lesson 1 alongside final two-sentence resolutions.
What did I learn?	Learners established that the inner structure of an atom — specifically the number of electrons in the outer shell — determines how many electrons an atom loses, gains, or shares, producing a specific ion charge called valency; that valency values are cross-multiplied to produce the only chemically correct formula for a compound; and that the formulae $\text{Ca}(\text{NO}_3)_2$ for CAN and $(\text{NH}_4)_2\text{HPO}_4$ for DAP are therefore not arbitrary labels but necessary consequences of the electron arrangements of calcium, nitrogen, hydrogen, oxygen, and phosphorus.
How does this explain the phenomenon?	Learners explained that: a tea farmer in Kericho sees different formulae on her fertiliser bags because calcium (Group 2, arrangement 2,8,8,2) always forms Ca^{2+} while ammonium carries a 1+ charge and nitrate a 1- charge and phosphate a 3- charge; cross-multiplying these valencies produces the fixed, unalterable ratios encoded in the formulae; balanced equations verify that no atoms are created or destroyed in these fixed-ratio combinations; and therefore the specific formula on each bag is a direct, readable record of the electron structure of the atoms inside — making the periodic table the key to understanding Kenyan agriculture at the molecular level.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.